

ABELE 2025

AGRICULTURAL & BIOSYSTEMS MACHINERY & MECHANIZATION

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Which of the following countries follows the low land area to farmer ratio model, focusing on mechanization to increase yields and cropping intensities?

- a. USA
 - b. Japan
 - c. Thailand
 - d. Philippines
- 



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Which country does
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- a. Philippines
 - b. Taiwan
 - c. South Korea
 - d. USA
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- a. Philippines
- b. Taiwan
- c. South Korea
- d. USA

1. High land area to farmer ratio (USA Model)

- Motivated into increasing the level of mechanization in order to cultivate large agricultural lands with limited available manpower and take advantage of favorable agricultural commodity prices.
- Thailand, Malaysia and Indonesia follow the USA model

2. Low land area to farmer ratio (Japanese Model)

- Motivated into increasing the level of mechanization in order to increase yields and cropping intensities to meet the growing demands for food and agricultural raw materials.
- South Korea, China, Taiwan, Sri Lanka and the Philippines follow the Japanese model.



What is considered the core of agricultural machinery manufacturing, marketing, and after-sales service?

- a. PCARRD
- b. AMMDA
- c. AMDP
- d. BPRE



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Which organization serves as the Philippine counterpart of the Regional Network for Agricultural Machinery (RNAM) and conducts research and extension on agricultural mechanization?

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Government Sector

Department of Agriculture (DA)



- **Agricultural and Fisheries Mechanization Committee (AFMeC)** – Advisory on mechanization concerns
- **Bureau of Postharvest Research and Extension (BPRE)** – Post-harvest and corn mechanization
- **Philippine Rice Research Institute (PhilRice)** – Rice mechanization
- **Bureau of Soils and Water Management (BSWM)** – Small-scale irrigation projects
- **National Irrigation Administration (NIA)** – Large-scale irrigation projects
- **Bureau of Plant Industry (BPI)** – Vegetable mechanization
- **National Food Authority (NFA)** – Grain storage and marketing
- **Bureau of Agricultural Research (BAR)** – Agricultural Engineering Research, Development, and Extension (RDE) Network
- **Regional Agricultural Engineering Groups (RAEG)** – Regional agricultural mechanization

University of the Philippines Los Baños

- **Agricultural Mechanization Development Program (AMDP)** – Agricultural mechanization research (Regional Network for Agricultural Mechanization counterpart)
- **Agricultural Machinery Testing and Evaluation Center (AMTEC)** – Agricultural machinery testing and standards

Central Luzon State University

- **Philippine-Sino Center for Agricultural Technology (PHILSCAT)** – Promotion of Chinese agricultural machinery

Department of Science and Technology

- **Philippine Council for Agriculture, Forestry and Natural Resources Research and Development (PCARRD)** – Policy direction for agricultural mechanization

Professional Regulation Commission

- **Board of Agricultural Engineering (BOAE)** – Regulates the agricultural engineering profession

Private Sector

- **Philippine Society of Agricultural Engineers (PSAE)** – Organization of agricultural engineers
- **Agricultural Machinery Manufacturers and Distributors Association (AMMDA)** – Core of machinery manufacturing, marketing, and after-sales service



The introduction of machinery, such as tractors and harvesters, which replaced manual labor and increased productivity.

- a. Agriculture 1.0
- b. Agriculture 2.0
- c. Agriculture 3.0
- d. Agriculture 4.0



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It represents the ongoing digital revolution in farming. It is characterized by the convergence of technologies such as the Internet of Things (IoT), artificial intelligence (AI), robotics, and big data analytics.

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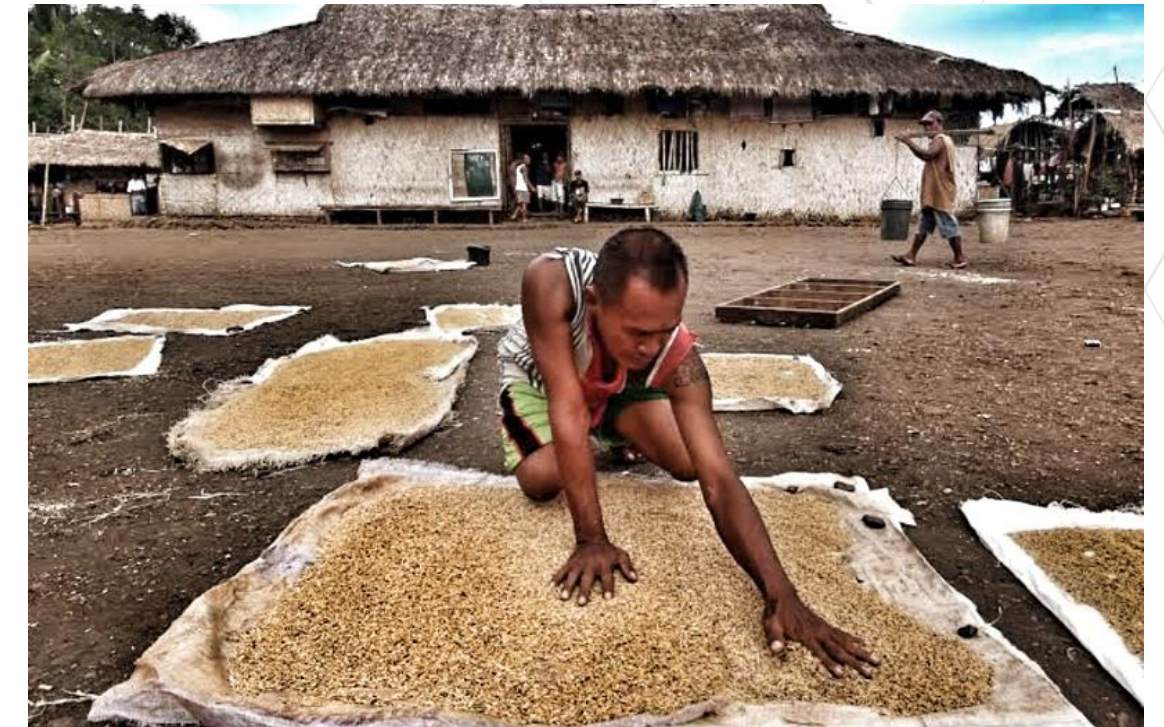
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Evolution of Agriculture

Agriculture 1.0 – The Era of Manual Labor

- Characterized by **simple tools, manual and animal labor**.
- Strong reliance on natural elements like the sun and rain.
- Farmers used **knowledge of the land, weather patterns, and traditional farming practices** passed down through generations.
- Dominated by subsistence farming—producing just enough food for the family's needs.



Agriculture 2.0 – The Era of Mechanization (1950s)

- Also called the “**Green Revolution**” due to a large increase in crop yield and productivity.
- Triggered by the Industrial Revolution and post–World War II industrial adaptation from military to civilian use.
- **Introduction of machinery** such as tractors and harvesters, replacing manual labor.
- Use of **chemical fertilizers and pesticides** to improve yields, though with negative environmental impacts.
- **Breeding programs** expanded through public universities, research institutes, and private companies.



Evolution of Agriculture

Agriculture 3.0 – The Precision Agriculture Era (1990s)

- Also known as **precision agriculture**.
- Used **GPS, remote sensing, and Geographic Information Systems (GIS)** to gather data on soil, crops, and weather.
- Created precise maps to apply fertilizers, pesticides, and irrigation **only where and when needed**.
- Reduced waste and minimized environmental damage.
- Public awareness grew about the environmental impacts of heavy fertilizer and chemical use.



Agriculture 4.0 – The Digital Revolution in Farming (Today)

- Driven by technologies like **Internet of Things (IoT), artificial intelligence (AI), robotics, and big data analytics**.
- Increased farm automation—robots handle planting, de-weeding, and harvesting.
- IoT sensors monitor soil moisture, crop conditions, and livestock health.
- AI analyzes data to optimize decisions and predict outcomes.
- Precision resource application reduces waste and environmental impact.
- Breeding programs develop varieties with lower input needs and greater resilience to environmental stress.





Any soil manipulation which changes the structure of the soil, kills weeds and rearranges dead plant materials.

- a. Soil arrangements
- b. Tillage
- c. Soil action
- d. Mechanization



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You are managing a production facility and need to choose between one large machine and two smaller machines for your operations. Which option is more advantageous if you want to minimize total stoppage and ensure continued production in case of equipment failure?

- a. One large machine
- b. Two smaller machines
- c. One large machine with backup
- d. No preference



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A farmer needs to prepare a new field that hasn't been cultivated in years. The soil is compacted and needs deep cutting and loosening to improve aeration and water absorption. Which type of tillage and implement should the farmer use?

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- a. Primary Tillage
 - b. Secondary Tillage
 - c. General-purpose tillage
 - d. Spike-tooth harrows



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Secondary tillage
prepare the soil up to
what to depth?

- 
- a. Up to 3"
 - b. 3" to 6"
 - c. 6" to 36"
 - d. Greater than 36"



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TILLAGE TYPES

1. Primary Tillage:

- Involves initial cutting, breaking, and inversion of the soil.
- **Implements:** Moldboard, disc, chisel plows, and subsoilers.
- **Depth:** 6 to 36 inches.
- Commonly referred to as **plowing**.

2. Secondary Tillage:

- Focuses on further breaking, pulverizing, and leveling the soil to prepare it for planting.
- **Implements:** Disc harrows, spike-tooth harrows, and spring-tooth harrows.
- **Depth:** 3 to 6 inches.
- Commonly referred to as **harrowing**.

3. General-Purpose Tillage:

- Combines primary and secondary tillage in a single operation.
- **Implements:** Rotavators and floating tillers.
- **Depth:** Up to 6 inches.
- Commonly referred to as **rotavating/rototilling**.



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-----.

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- a. Speed
 - b. Disc diameter
 - c. Depth of cut
 - d. Disc angle



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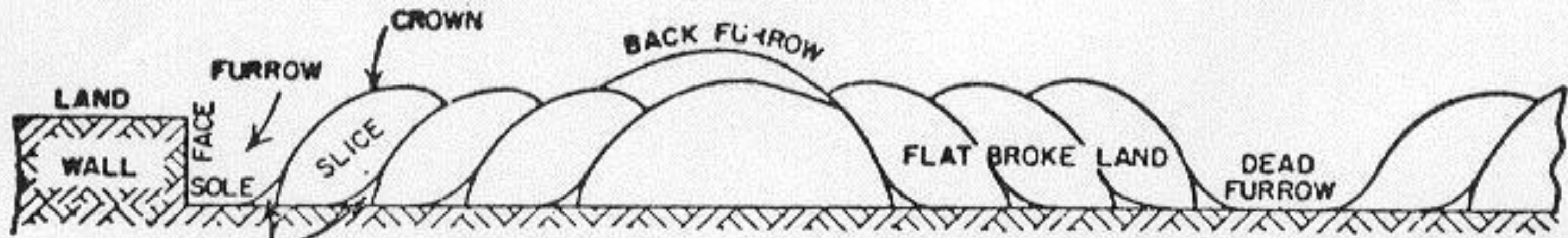
Refers to the trench left equal to two furrows when furrowslices are thrown on opposite sides.

- a. Backfurrow
- b. Deadfurrow
- c. Furrowslice
- d. Furrow wall

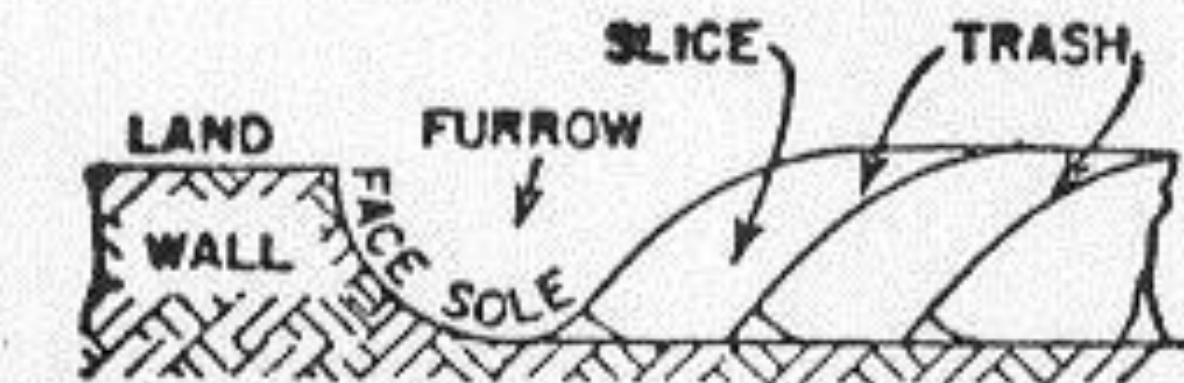


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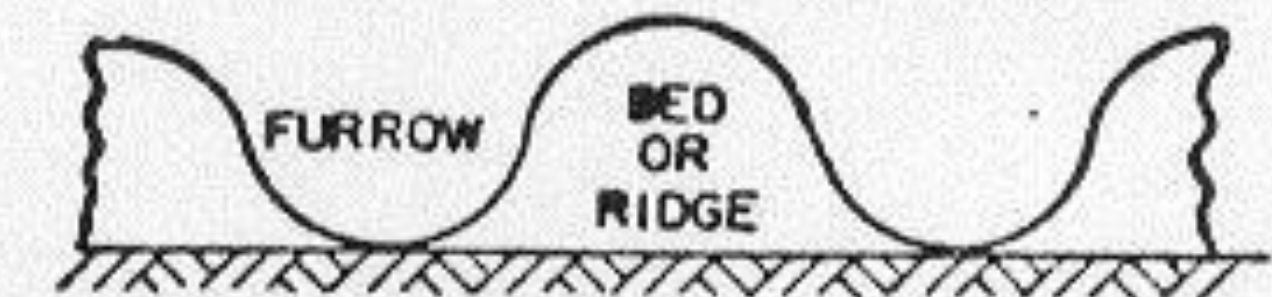
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A-FURROWS MADE BY MOLD BOARD PLOW, KINDS AND PARTS OF A FURROW



B-FURROW MADE BY DISK PLOW



C-FURROWS MADE BY MIDDLE BREAKER OR LISTER



Plowing Terms



1. **Land** – unplowed soil.
2. **Furrow slice** – soil cut, lifted, inverted and thrown to one side of the plow bottom.
3. **Furrow** – trench or canal left by the furrowslice.
4. **Backfurrow** – raised ridged when two furrowslices overlap each other.
5. **Deadfurrow** – trench left equal to two furrows when furrowslices are thrown on opposite sides.
6. **Furrow wall** – unbroken side of the furrow.



It refers to the central part where the various components of the plow are attached.

- 
- a. Tail piece
 - b. Frog
 - c. Plow bottom
 - d. Share



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It cuts the soil to a depth of 6" to 18 ".

- 
- a. Moldboard plow
 - b. Disc plow
 - c. Rotavator
 - d. Disc harrow



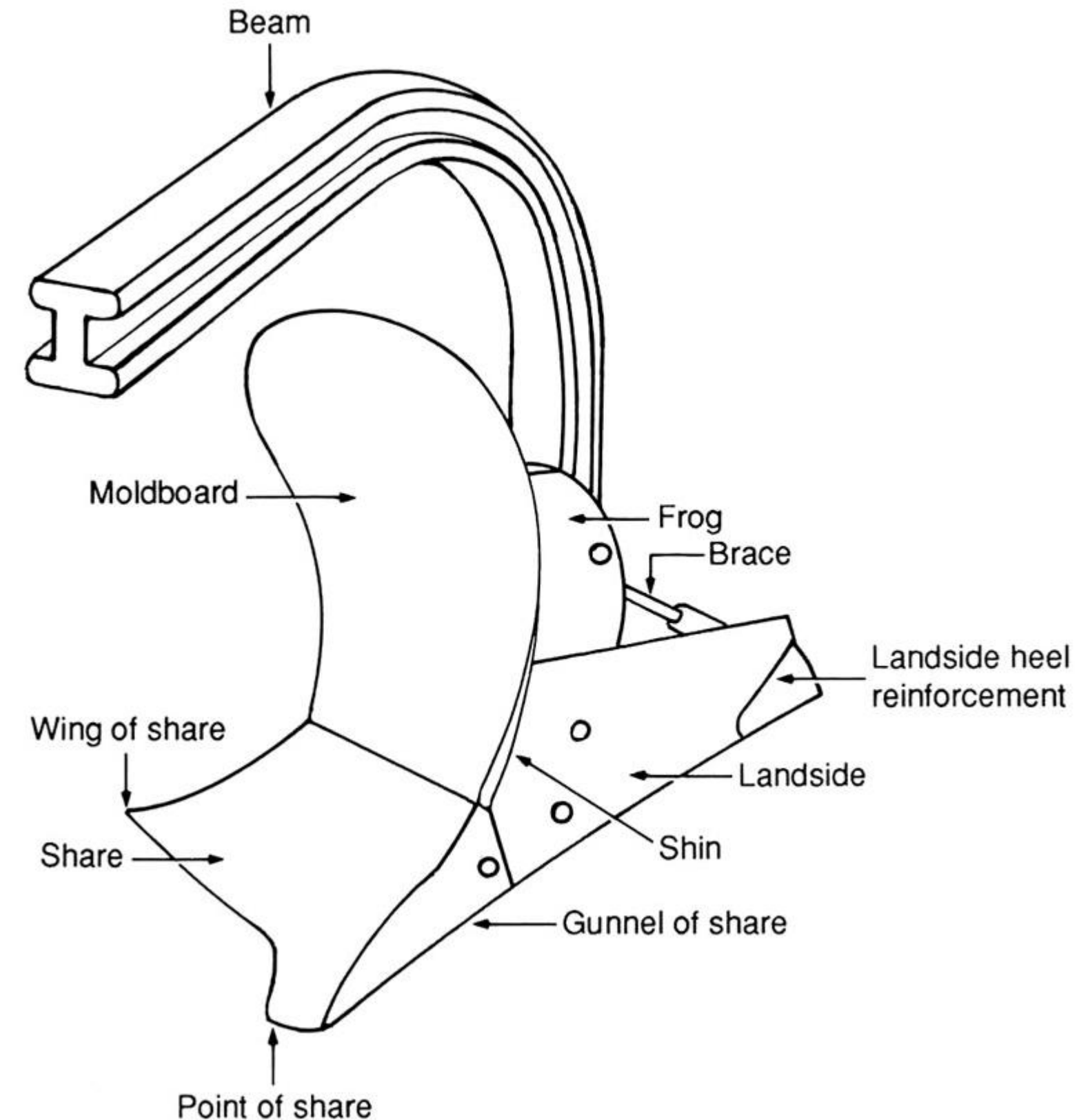
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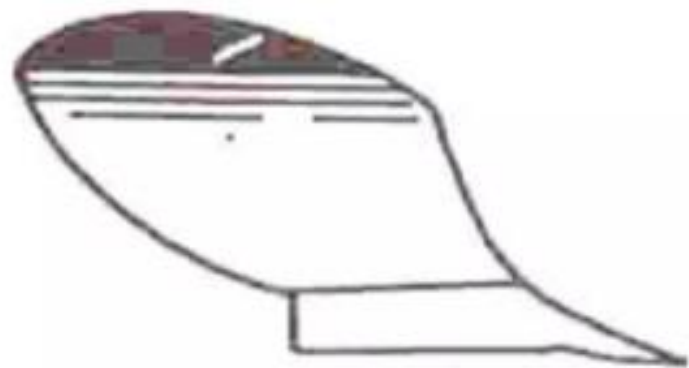
Moldboard plow - one of the oldest and most important agricultural equipment for land preparation. It cuts the soil to a depth of 6" to 18 ".

Part of moldboard plow bottom

1. **Share** – provides the cutting edge of the plow bottom.
2. **Moldboard** – receives the furrowslice from the share; lifts, inverts and throws it to one side of the plow bottom.
3. **Landside** – counteracts the side pressure exerted by the furrowslice on the plow bottom.



Types of mould board



GENERAL PURPOSE
MOULD BOARD



STUBBLE MOULDBOARD



SOD OR BREAKER MOULD BOARD



SLAT MOULDBOARD





While adjusting a moldboard plow, a farmer bends the point of the share outward toward the unplowed land to ensure it cuts the proper width. Which clearance is being adjusted?

- a. Vertical clearance
- b. Horizontal clearance
- c. Depth clearance
- d. Share clearance



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Clearances of the moldboard plow bottom:

1. Vertical clearance

– bend downward of the point of the share to make the plow penetrate the soil to the proper depth.

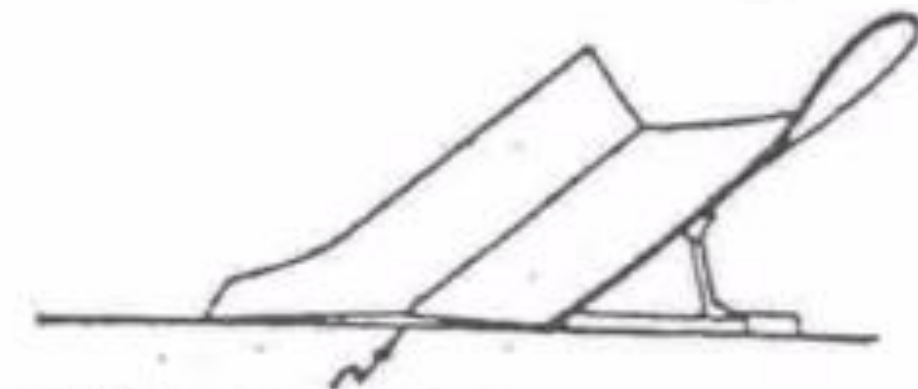
1. Horizontal clearance

– bend outward or sideward of the point of the share towards the unplowed land to make the plow cut the proper width.



CLEARANCE - 3 to 5 mm

VERTICAL SUCTION



SIDE CLEARANCE - 5 mm

HORIZONTAL SUCTION



A farm technician wants to determine the size of a moldboard plow bottom. Which measurement should he take?

- a. Perpendicular distance between the wing and the point of the share
- b. Horizontal distance between the wing and the point of the share
- c. Perpendicular distance between the moldboard and the point of the share
- d. Horizontal distance between the moldboard and the wing



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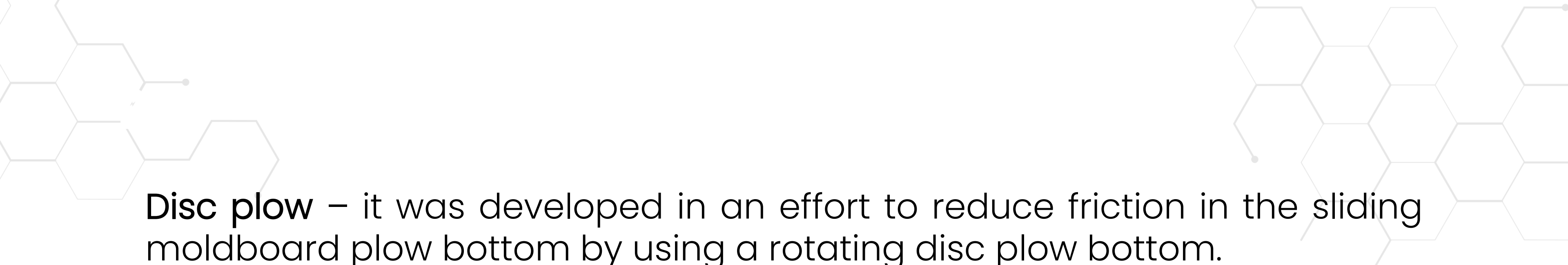
A farmer switches from a sliding moldboard plow to a plow with a rotating bottom to reduce friction during operation. Which implement is he using?

- a. Disc plow
- b. Rotary plow
- c. Disc harrow
- d. Circular moldboard plow



A farmer switches from a sliding moldboard plow to a plow with a rotating bottom to reduce friction during operation. Which implement is he using?

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Disc plow – it was developed in an effort to reduce friction in the sliding moldboard plow bottom by using a rotating disc plow bottom.

Basic parts of disc plow bottom:

- a) **Disc** – round and concave disc of heat-hardened steel with sharpened edges to cut the soil. It is rotated by the soil during soil cutting.
- b) **Scraper** – curved plate placed on the concave side of the disc to scrape off the soil and throw it on one side of the plow bottom.
- c) **Rear furrow wheel** – rear wheel that presses against the furrow wall to resist the side pressure exerted by the soil against the plow.



What is the range of degrees for the tilt angle that allows a disc to penetrate the soil to the proper depth?

- 
- a. 10–15 degrees
 - b. 15–25 degrees
 - c. 25–35 degrees
 - d. 0–5 degrees



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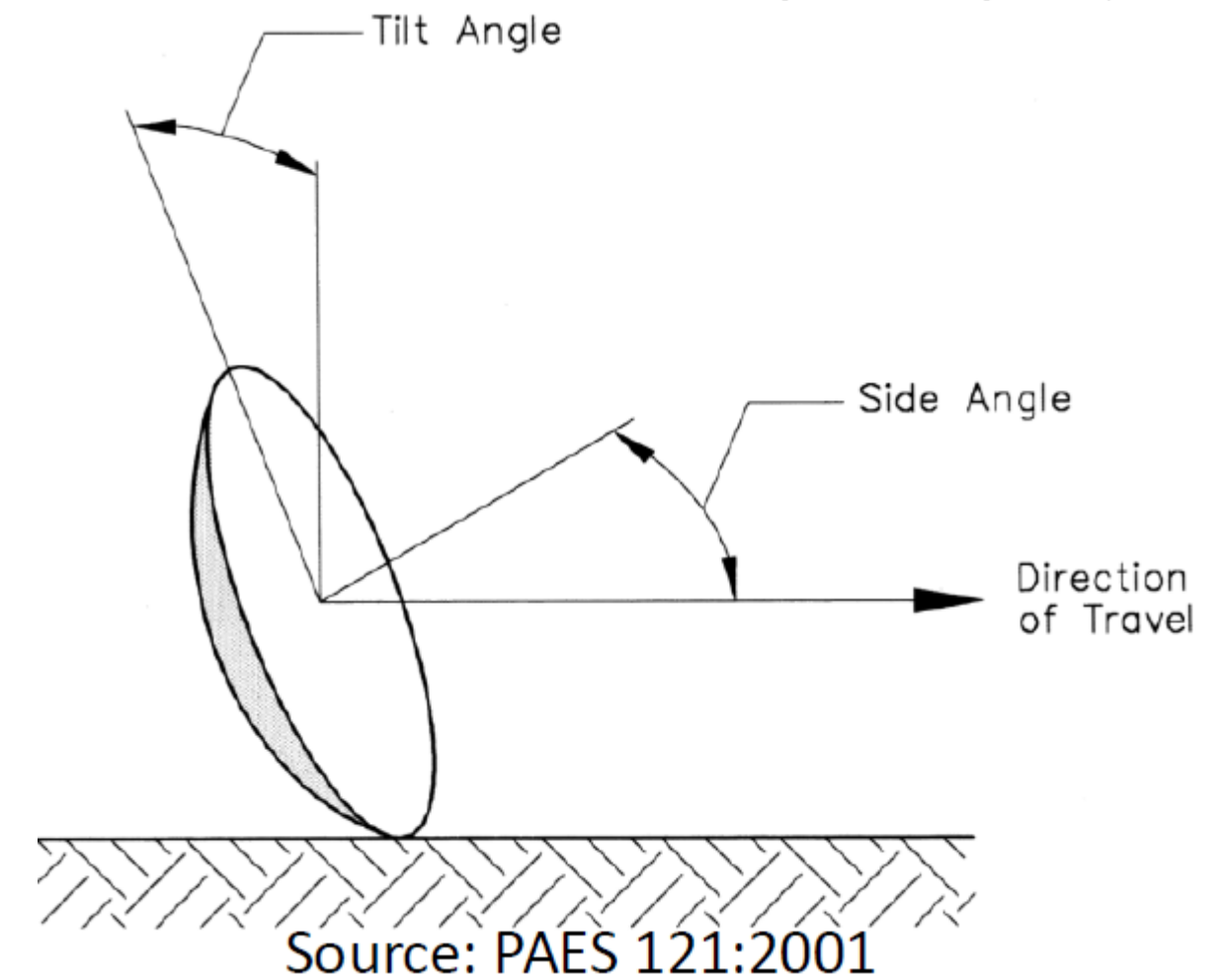
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ANGLES OF THE DISC PLOW

Tilt angle – angle the disc makes with the vertical to make the disc penetrate the soil to the **proper depth** (15 – 25 degrees).

Disc angle – angle the disc makes with the direction of travel to make the disc cut the **proper width** and allow the disc to rotate (42 – 45 degrees).



What is the required force to pull a single moldboard plow on a clay loam soil having 30 cm and 15 cm width and depth of cut, respectively. Soil draft is 8 psi and the average speed is 5 kph.

- a. 4 hp
- b. 4.6 hp
- c. 5 hp
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What is the required force to pull a single moldboard plow on a clay loam soil having 30 cm and 15 cm width and depth of cut, respectively. Soil draft is 8 psi and the average speed is 5 kph.

$$F = 8 \frac{\text{lb}}{\text{in}^2} \times \left(\frac{1 \text{ kg}}{2.2 \text{ lb}} \right) (30 \times 15) \text{cm}^2 \times \left(\frac{1 \text{ in}}{2.54 \text{ cm}} \right)^2$$
$$F = 253.64 \text{ kg}$$

$$1 \text{ kg} = 2.2 \text{ lb}$$
$$1 \text{ hp} = 76.2 \text{ kg-m/sec}$$

$$\text{DHP} = F \times S$$

$$\text{DHP} = (253.64 \text{ kg}) \left(\frac{5 \text{ km}}{\text{hr}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ hr}}{3600 \text{ sec}} \right) \left(\frac{1 \text{ hp}}{76.2 \frac{\text{kg-m}}{\text{sec}}} \right)$$

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A power tiller was tested on a 10-m wide plot. During the test, machine made 22 rounds to complete the plowing operation using two 0.3 m-diameter plow. What is the average swath of power tiller?

a. 0.23 m b. 0.37 m c. 0.42 m d. 0.51 m

Compute for the disc angle of plow.

a. 40° b. 50° c. 60° d. 70°



$$\text{Swath} = \frac{\text{Width of plot}}{2n}$$

where n = number of rounds

$$\text{Disc Angle} = \sin^{-1} \left(\frac{\text{swath}}{\text{diameter}} \right)$$

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$$\text{Swath} = \frac{\text{Width of plot}}{2n}$$

$$\text{Swath} = \frac{10 \text{ m}}{2(22)}$$

$$\text{Swath} = 0.23 \text{ m}$$



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$$\text{Disc Angle} = 50^\circ$$

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Compute for the disc angle of plow.

a. 40° b. 50° c. 60° d. 70°



The average swath of the power tiller was found to be 0.56 m. During the test, the machine made 12 rounds to complete the plowing operation using two plows, each with a 0.7 m diameter, set at a disc angle of 60 degrees. Determine the width of the plot where the test was conducted.

- a. 0.65 m; 13.44 m
- b. 0.75 m; 14.34 m
- c. 0.87 m; 10.55 m
- d. 0.90 m; 11.08 m

$$\text{Swath} = \frac{\text{Width of plot}}{2n}$$

$$0.56 \text{ m} = \frac{W}{2(12)}$$

$$W = 13.44 \text{ m}$$

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Disc Angle =

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$$60^\circ = \sin^{-1} \left(\frac{0.56 \text{ m}}{\text{diameter}} \right)$$

$$\text{Diameter} = 0.65 \text{ m}$$

$$\text{Swath} = \frac{\text{Width of plot}}{2n}$$

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Determine the hectares plowed when a tractor is operating at 5.7 kph for 6 hours and is pulling four 42-cm moldboard bottoms at a depth of 15 cm if field efficiency is 82%.

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- c. 4.71 ha
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$$C = SWEff/10$$

$$C = \frac{(5.7 \text{ kph})(4 \times 42 \text{ cm})(0.82)}{10}$$

$$C = 0.7852 \text{ ha/hr}$$

$$\text{Plowed area} = 0.7852 \text{ ha/hr} \times 6 \text{ hr}$$

$$\text{Plowed area} = 4.71 \text{ ha}$$

$$C = SWEff/10$$

S (Speed) = kph

W (Width of cut) = m

Eff (Field efficiency) = decimal

Effective width of a cut of a single-bottom moldboard plow drawn by a power tiller is 10 cm. If machine is used to plow 2ha farm, what would be the distance traveled?

- a. 20,000 m
- b. 200 km
- c. 20 km
- d. 2 km

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Area = Distance x Effective width

$$2 \text{ ha} \times \frac{10,000 \text{ m}^2}{1 \text{ ha}} = D \times 0.10 \text{ m}$$

$$D = 200,000 \text{ m or } 200 \text{ km}$$



Type of plowing where field is plowed in continuous circuits, starting either the center of the field or from the outer edge.

- a. Return plowing
- b. Round-about plowing
- c. Consecutive plowing
- d. One-way plowing



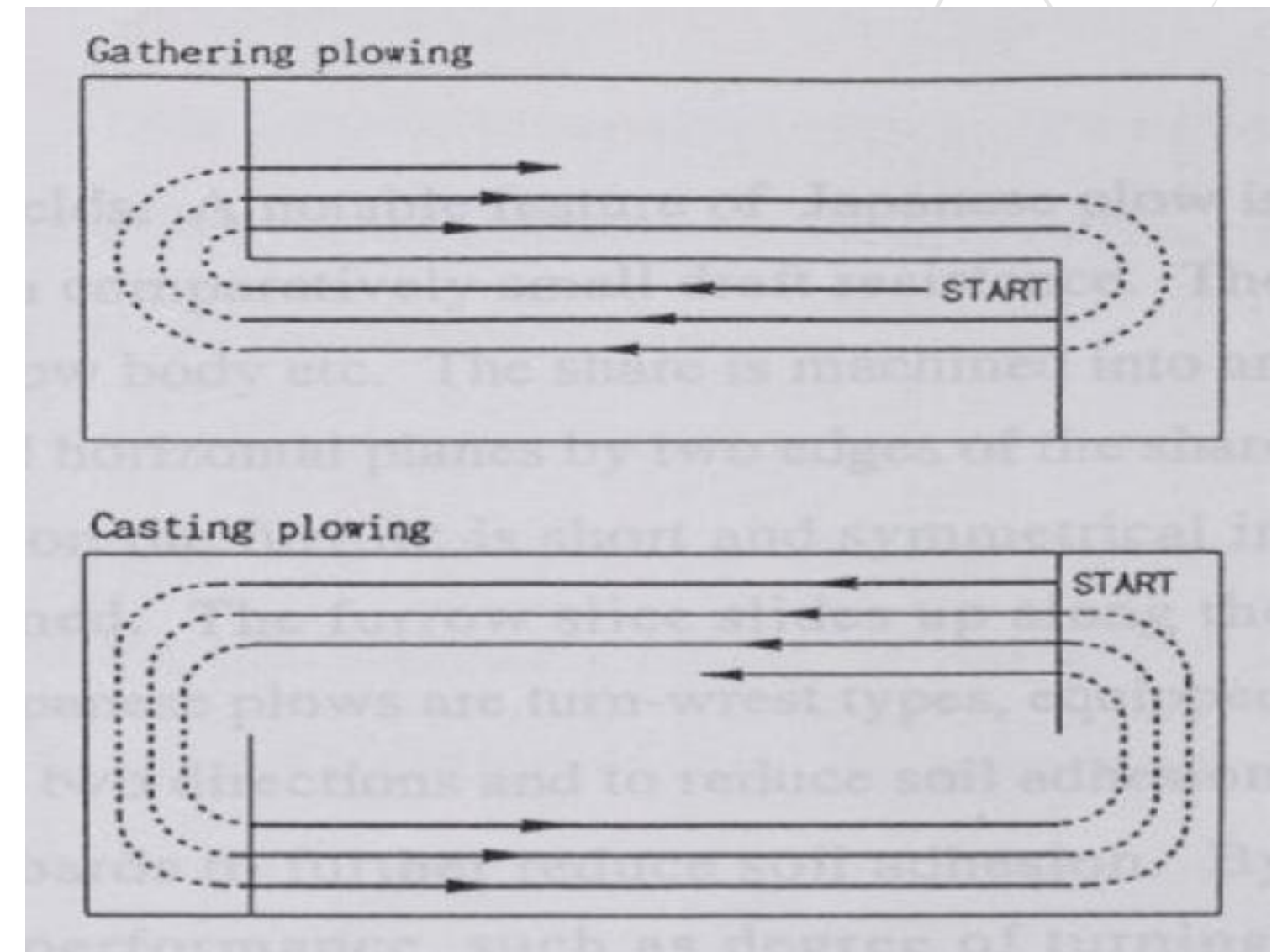
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METHODS OF PLOWING

1. **Return plowing** – moves two-way in a longitudinal direction. Gathering plowing and casting plowing are included in this method.

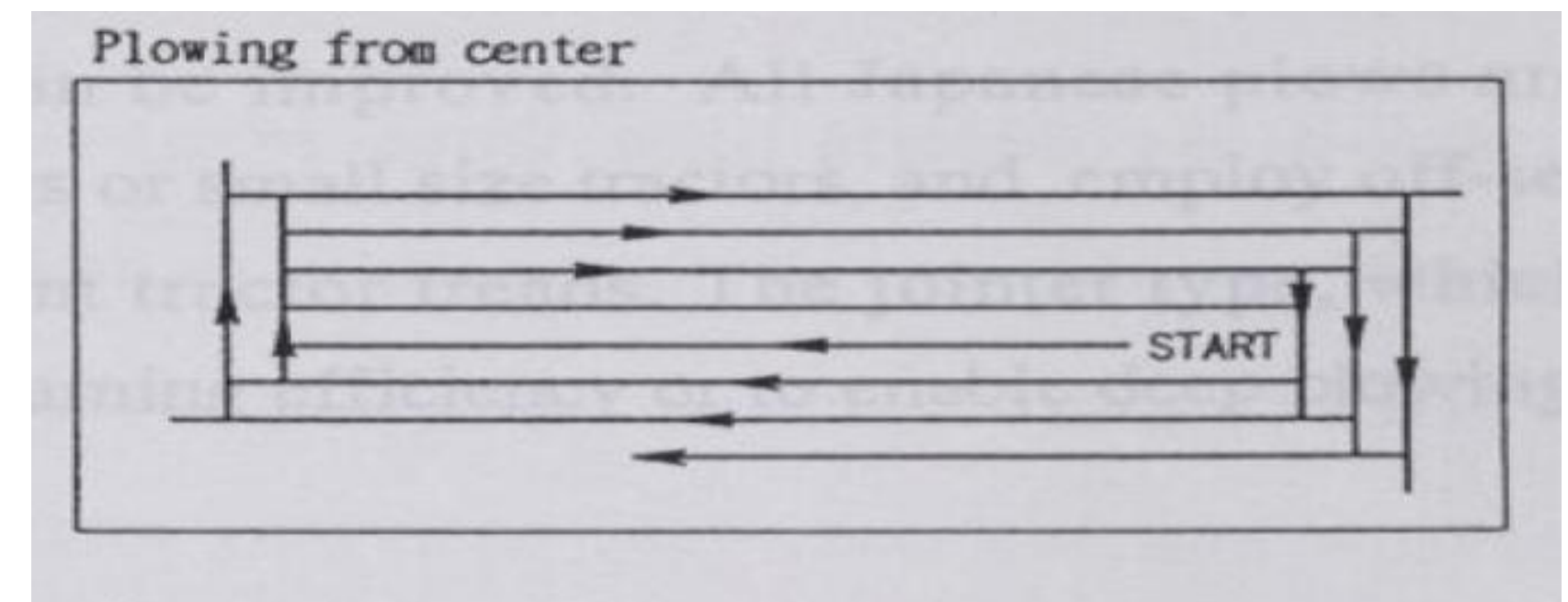
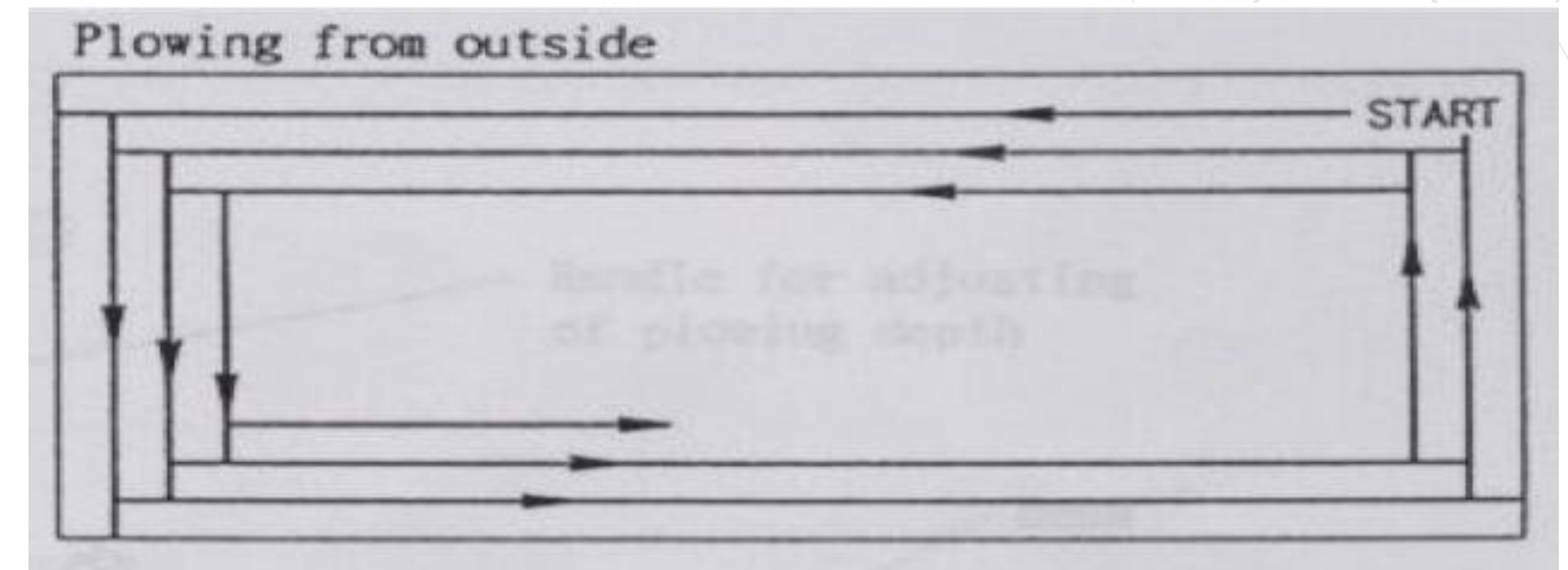
- **Gathering plowing** – plowing starts at the center line of the field
- **Casting plowing** – plowing starts from the outer side of the field



METHODS OF PLOWING

2. **Round-about plowing** – field is plowed in continuous circuits, starting from either the center of the field or from the outer edge.

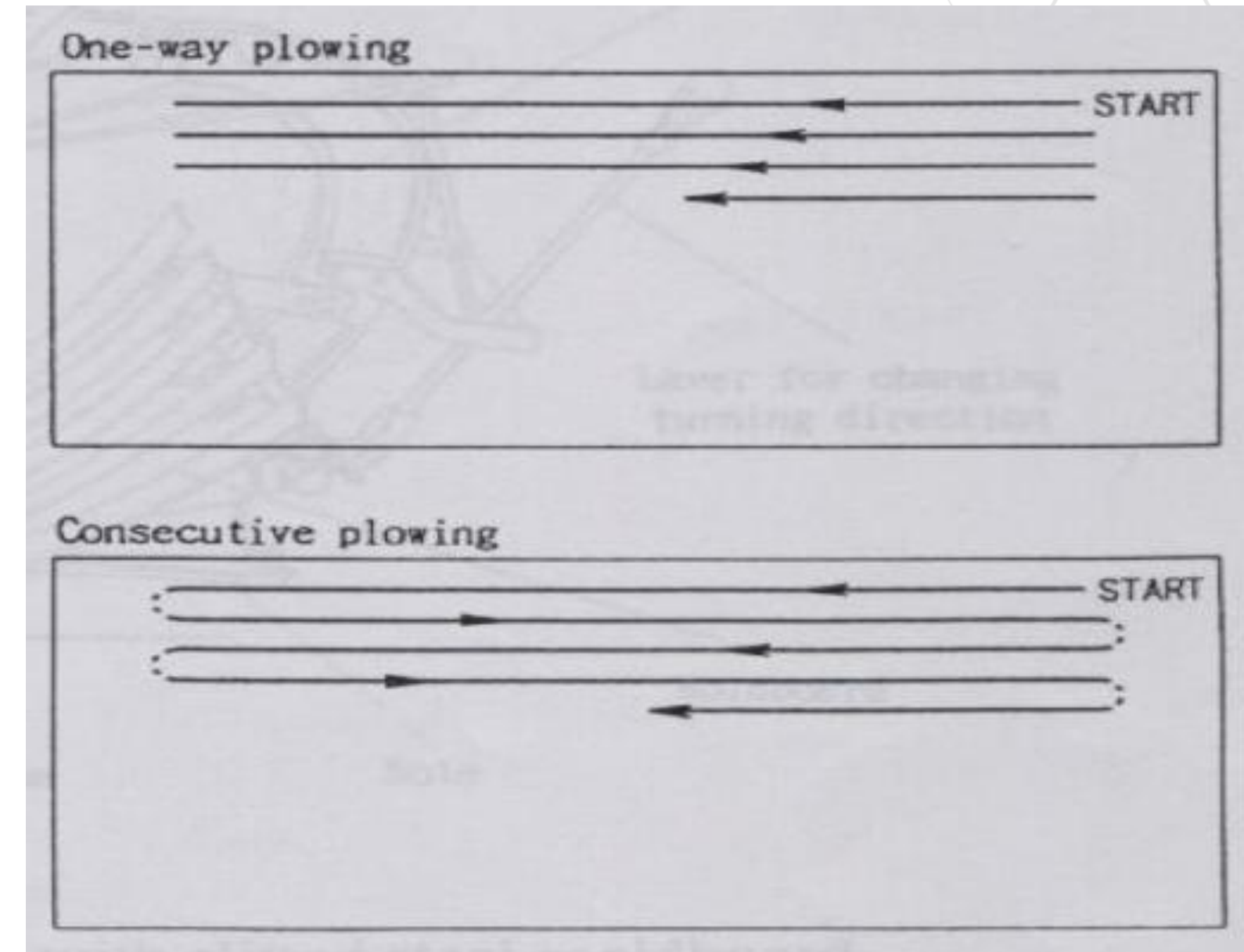
- **Plowing from outside** – plowing is at the corner field, and the circuit proceeds inwards, turning the slice toward the outer side.
- **Plowing from center** – a small square is first plowed in the same way as gathering plowing, then circuits proceed outwards, turning the furrow slice toward the center.



METHODS OF PLOWING

3. **One-way plowing** – the land is plowed in one direction only. This procedure is suitable for hillside plowing.

4. **Consecutive plowing** – the land is plowed in both directions side by side. This procedure can be practiced by reversible type, turn-wrest type, or symmetric tillage implements such as a two-way type driven disk plow.





In a disc harrow, which component is made up of a number of discs mounted on a common axle?

- a. Disc gang
- b. Disc blade
- c. Harrow frame
- d. Gang bolt



In a disc harrow, which component is made up of a number of discs mounted on a common axle?

- a. Disc gang
- b. Disc blade
- c. Harrow frame
- d. Gang bolt



A farmer is using a disc harrow with four gangs of discs—two gangs in front arranged like a single-action harrow, and two more gangs at the rear. What type of disc harrow is this?

- a. Single-action disc harrow
- b. Double-action disc harrow
- c. Offset disc harrow
- d. Tandem disc harrow



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Secondary Tillage Equipment

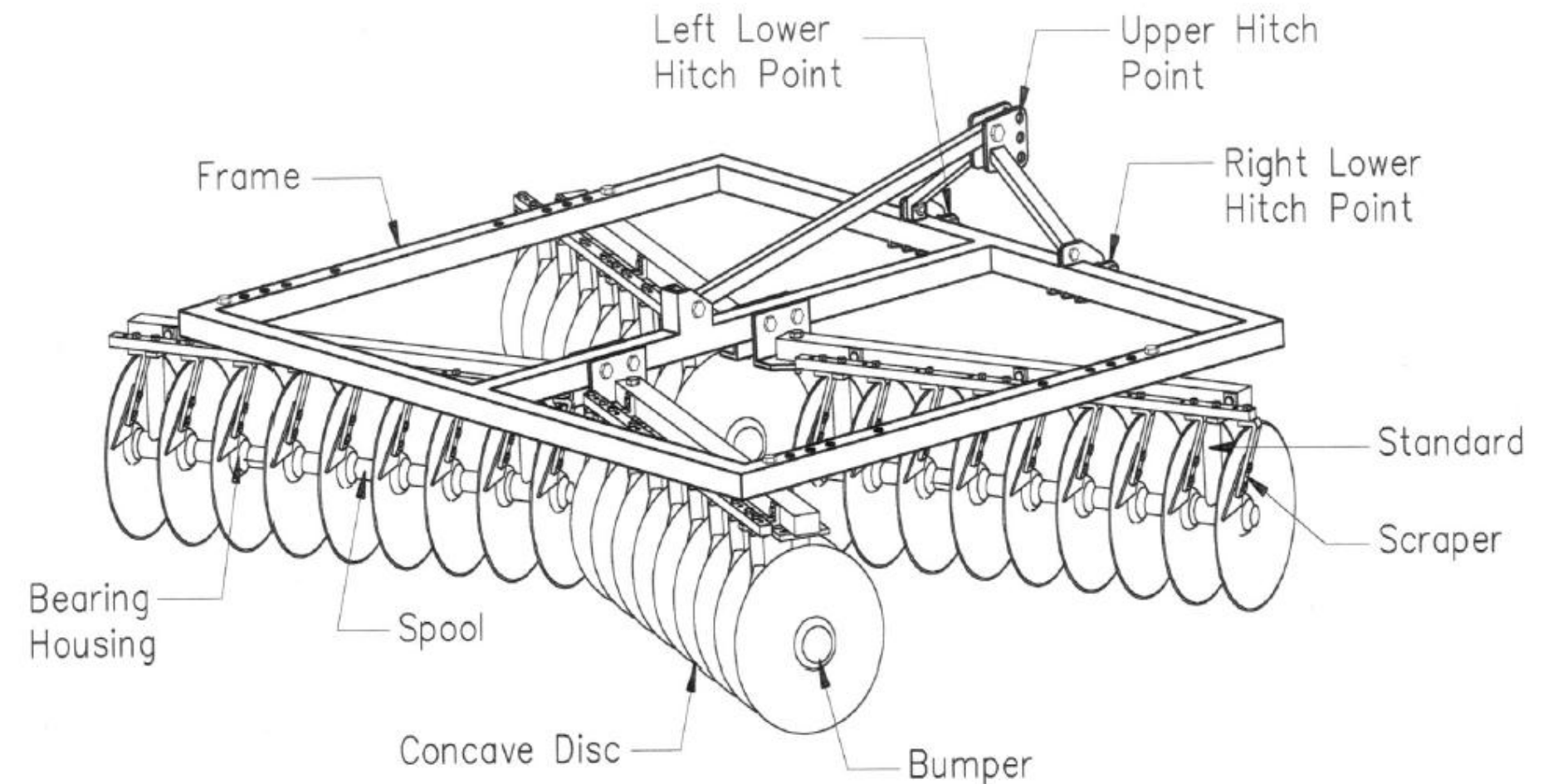
1. **Disc harrow** – consists of two or more sets of disc gangs cutting the soil to a depth of 3" to 6". A disc gang is made up of a number of discs mounted on a common axle.

Types of disc harrows

- a) **Single-action disc harrow** – consists of two gangs of discs placed end-to-end which throws the soil in opposite directions.
- b) **Double-action disc harrow** – consists of four gangs of discs. Two gangs in front as in single –action and two gangs at the rear.
- c) **Offset disc harrow** – consists of two gangs of disc. One gang located behind the other and operated in an offset position in relation to the tractor.

Basic parts of disc harrow

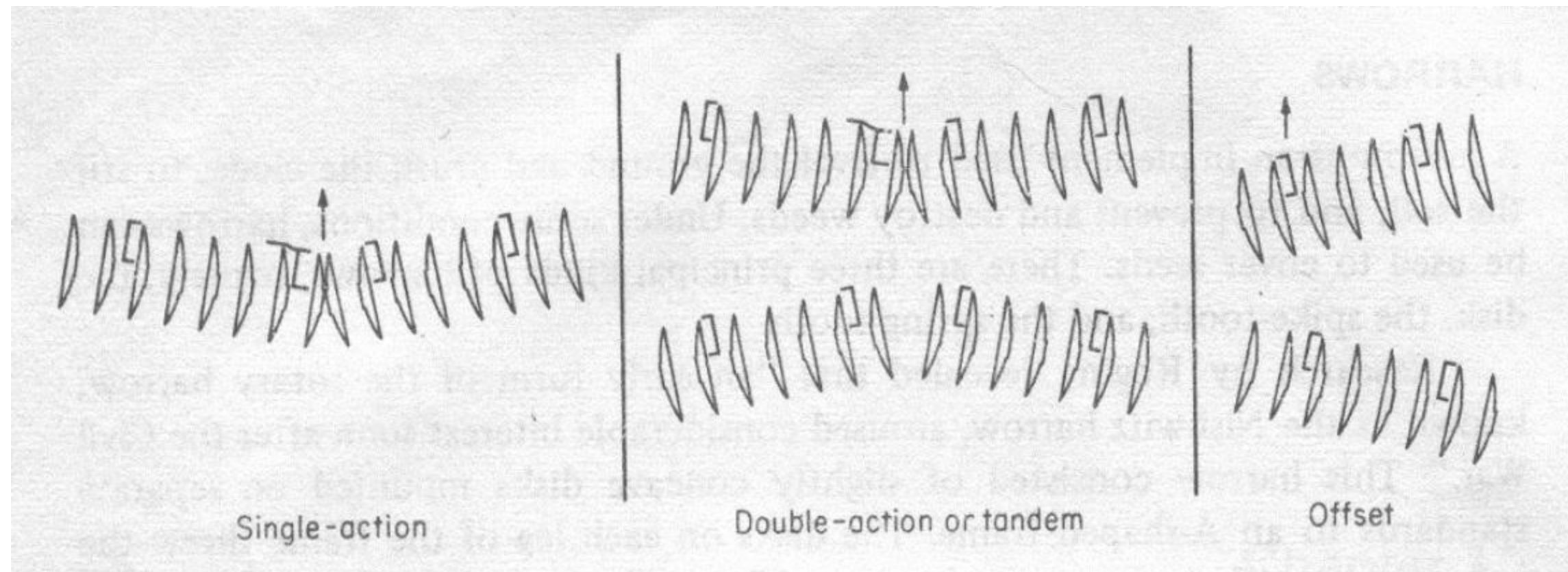
- a) **Discs** – round and concave discs
- b) **Disc gang** – an assembly of discs all rotating together on a common shaft (3 – 13 discs/gang at 6" – 12 " spacing per disc).
- c) **Bumpers** – a washer on one of the common shaft to absorb the end thrust of the gang.
- d) **Bearings** – two to three bearings per gang.
- e) **Spool** – flanged tube inserted on the common shaft between two discs to maintain disc spacing
- f) **Bumper** – a washer on one of the common shaft to absorb the end thrust of the gang
- g) **Scraper** – straight or curved plate placed on the concave side of the disc to scrape off the soil



Source: PAES 120:2001

Types of disc harrows

- a) **Single-action disc harrow** – consists of two gangs of discs placed end-to-end which throws the soil in opposite directions.
- b) **Double-action disc harrow** – consists of four gangs of discs. Two gangs in front as in single –action and two gangs at the rear.
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It consists of long rigid spikes clamped or welded to crossbars in a staggered arrangement for maximum stirring and raking of the soil, with crossbars that can be rotated to change the angle of the teeth.

- a. Spike-tooth harrow
- b. Spring-tooth harrow
- c. Tine harrow
- d. Drag harrow



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- a. Spike-tooth harrow
- b. Spring-tooth harrow
- c. Tine harrow
- d. Drag harrow



Secondary Tillage Equipment



2. Spike-tooth harrow – consists of long rigid spikes clamped or welded to cross bars on a staggered arrangement to attain maximum stirring and raking of the soil. The cross bars can be rotated to change the angle of the teeth

3. Spring-tooth harrow – consists of long, flat and curved teeth made from spring steel. The curved teeth are welded to cross bars on a staggered arrangement. The spring characteristic allows the teeth to flex and slide over obstructions.

General-purpose tillage equipment:

1. Rotavators – constructed with a set of cutting knives or tines mounted on a horizontal power-driven shaft rotating at high speed. The knives slice thin sections of the soil and pulverize them by centrifugal force.

- Rotavators are mounted on 2-wheel or 4-wheel tractors and driven by the tractor PTO.
- Rotavators are suited for both dry and wet land operations.

2. Floating power tillers – specially designed power tillers for wet land operation fitted with front-mounted cutting blades on a cagewheel



What is the cutting width of single-disk harrow set at 18 degrees working angle containing 20 disks spaced at 30 cm? Disk diameter is 40 cm.

- a. 5.82 m
- b. 6.18 m
- c. 7.05 m
- d. 8.89 m



Tandem type: $W = 0.95NS + 1.2D$

Offset type: $W = 0.95NS + 0.6D$

Single-action: $W = 0.95NS + 0.3D$

Double offset: $W = 0.95NS + 0.85D$

N = number of disks

S = Spacing

D = disk diameter



What is the cutting width of single-disk harrow set at 18 degrees working angle containing 20 disks spaced at 30 cm? Disk diameter is 40 cm.

- a. 5.82 m
- b. 6.18 m
- c. 7.05 m
- d. 8.89 m

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- a. 5.82 m
- b. 6.18 m
- c. 7.05 m
- d. 8.89 m

$$W = 0.95NS + 0.3D$$

$$W = 0.95(20)(30) + 0.3(40)$$

$$W = 582 \text{ cm} = 5.82 \text{ m}$$



What is the cutting width of tandem-disk at 18 degrees working angle containing 20 disks spaced at 30 cm? Disk diameter is 40 cm.

- a. 5.82 m
- b. 6.18 m
- c. 7.05 m
- d. 8.89 m



What is the cutting width of tandem-disk at 18 degrees working angle containing 20 disks spaced at 30 cm? Disk diameter is 40 cm.

- a. 5.82 m
- b. 6.18 m**
- c. 7.05 m
- d. 8.89 m

$$W = 0.95NS + 1.2D$$

$$W = 0.95(20)(30) + 1.2(40)$$

$$W = 618 \text{ cm} = 6.18 \text{ m}$$

A 4-wheel tractor with 3x14 inches moldboard plow is to operate on clay-loam soil ($SD = 0.56\text{kg/cm}^2$) at a depth of 25.4cm. The maximum draft and percent increase in draft due to speed are given below. Determine the highest gear setting that can sustain the draft requirement for the operation.

Gear setting	Speed (kph)	Maximum draft (kg)	Increase in draft due to speed
1L	3.2	9,000	14
2L	4.8	3,100	28
3L	6.4	2,200	42
4L	8.0	1,300	56

a. 1L b. 2L c. 3L d. 4L




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2L	4.8	3,100	28
3L	6.4	2,200	42
4L	8.0	1,300	56

Specific Draft = Specific Draft x W x D

Specific Draft = $0.56\text{kg/cm}^2 \times (3 \times 14 \text{ inches} \times (2.54 \text{ cm/1 inch}) \times 25.4 \text{ cm}$

Specific Draft = 1517.42 kg

Adjusted Draft at 3L

Adjusted Draft = Specific Draft x Increase in Draft

Adjusted Draft = $1517.42 \text{ kg}(1.42) = 2, 154.74 \text{ kg}$

Use 2, 200 kg at 3L

A 4-wheel tractor with 3x14 inches moldboard plow is to operate on clay-loam soil ($SD = 0.56\text{kg/cm}^2$) at a depth of 25.4cm. The maximum draft and percent increase in draft due to speed are given below. Determine the highest gear setting that can sustain the draft requirement for the operation.

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4L	8.0	1,300	56

a. 1L b. 2L **c. 3L** d. 4L

A 4-wheel tractor with 3x14 inches moldboard plow is to operate on clay-loam soil ($SD = 0.56 \text{ kg/cm}^2$) at a depth of 25.4cm. The maximum draft and percent increase in draft due to speed are given below. Determine the tractor horsepower.

Gear setting	Speed (kph)	Maximum draft (kg)	Increase in draft due to speed
1L	3.2	9,000	14
2L	4.8	3,100	28
3L	6.4	2,200	42
4L	8.0	1,300	56

- a. 60 HP b. 62.9 HP c. 65 HP d. 67.12 HP




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1L	3.2	9,000	14
2L	4.8	3,100	28
3L	6.4	2,200	42
4L	8.0	1,300	56

Since we will use 3L Gear setting,
our S = 6.4 kph and Adjusted Draft = 2, 154.74 kg

$$\text{Draft} = \frac{DS}{274} \quad D = \text{kg}; S = \text{kph}$$

$$\text{Draft} = \frac{2154.74 \text{ kg} \times 6.4 \text{ kph}}{274} = 50.3297 \text{ HP}$$

$$\text{Tractor HP} = \text{Draft}/0.85 = 50.3297/0.85 = 59.21 \text{ HP}$$

A 4-wheel tractor with 3x14 inches moldboard plow is to operate on clay-loam soil ($SD = 0.56\text{kg/cm}^2$) at a depth of 25.4cm. The maximum draft and percent increase in draft due to speed are given below. Determine the tractor horsepower.

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4L	8.0	1,300	56

a. 60 HP

b. 62.9 HP

c. 65 HP

d. 67.12 HP



A four-wheel tractor with 3 – 36 cm moldboard plow is to operate on silty loam soil ($SD = 0.42/\text{cm}^2$) at a depth of 25 cm. The maximum drafts of the tractor at different working speeds are given below. Determine the tractor horsepower. Increase in draft at 3L = 135% and 4L = 149%.

GEAR SETTING	SPEED (KPH)	MAX. DRAFT (KG)
1L	2.4	4,000
2L	4.0	3,100
3L	5.6	2,200
4L	7.2	1,300
1H	9.6	1,000



A tillage action and transport operations utilized to loosen, load, carry, and unload soil

- 
- a. Land forming
 - b. Land grading
 - c. Land planing
 - d. Earthmoving



A tillage action and transport operations utilized to loosen, load, carry, and unload soil

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Tillage operation
which move soil to
create desired soil
configurations

- 
- a. Land forming
 - b. Land grading
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Primary tillage operation
which manipulates soil to
a greater depth than 300
mm

- 
- a. Deep tillage
 - b. Soil cultivation
 - c. Harrowing
 - d. Overall tillage



Primary tillage operation
which manipulates soil to
a greater depth than 300
mm

- 
- a. **Deep tillage**
 - b. Soil cultivation
 - c. Harrowing
 - d. Overall tillage



GENERAL TILLAGE TERMS AND KIND OF TILLAGE

broadcast tillage/overall tillage – tillage of an entire area as contrasted to a partial tillage as in bands or strips

deep tillage – primary tillage operation which manipulates soil to a greater depth than 300 mm

earthmoving – tillage action and transport operations utilized to loosen, load, carry, and unload soil

land forming – tillage operation which move soil to create desired soil configurations

land grading – tillage operation which move soil to establish a desired soil elevation and slope (leveling, contouring, cutting, and filling)

land planning – tillage operation that cuts and moves small layers of soil to provide smooth, refined surface condition



GENERAL TILLAGE TERMS AND KIND OF TILLAGE



broadcast tillage/overall tillage – tillage of an entire area as contrasted to a partial tillage as in bands or strips

oriented tillage – tillage operation which are oriented in specific paths or directions with respect to the sun, prevailing winds, previous tillage actions, or field base lines

rotary tillage – tillage operation employing power driven rotary action to cut, break up, and mix soil



Subsoiling under the plant row prior to planting usually intended for subsurface drainage

- 
- a. Optimum tillage
 - b. Subsoil tillage
 - c. Conservation tillage
 - d. Precision tillage



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Tillage system in which
only isolated bands of
soil is tilled

- 
- a. Band tillage
 - b. Strip tillage
 - c. Reduced tillage
 - d. Minimum tillage



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only isolated bands of
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- a. Band tillage
 - b. Strip tillage**
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 - d. Minimum tillage



In conservation tillage system, what is the minimum percentage of residue cover being maintained on the soil surface after planting ?

- a. 20%
 - b. 30%
 - c. 40%
 - d. 505
- 



In conservation tillage system, what is the minimum percentage of residue cover being maintained on the soil surface after planting ?

- a. 20%
 - b. 30%**
 - c. 40%
 - d. 505
- 



In tillage operation,
this is the reverse of middle
breaking or hilling up.

- 
- a. Incorporating
 - b. Mixing
 - c. Anchoring
 - d. Off barring



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TILLAGE OPERATION

1. **anchoring** – tillage to partially bury and thereby prevent movement of materials such as plant residues or artificial mulches
2. **bedding/ridging/listing** – tillage which forms a ridge and furrow soil configuration
3. **bulldozing** – pushing or rolling of soil by a steeply inclined blade
4. **Incorporating/mixing** – operation which mix or disperse foreign materials, such as pesticides, fertilizers or plant residues into the soil
5. **middlebreaking/hilling up** – operation wherein a lister is used in a manner that forms a furrow midway between two previous rows of plants
6. **off barring** – operation that cuts and throws the soil away from the base of plants.



Tillage tools which
are powered so that
some of their
movements are in
direction other than
along the line of
travel

- a. Complex
 - b. Dynamic
 - c. Multipowered
 - d. Simple
- 



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NOMENCLATURE FOR TILLAGE TOOLS AND IMPLEMENTS

1. **complex tillage tools** – tillage tools which rotate or move so that they present a varying boundary and contact area to the soil
2. **dynamic tillage tools** – tillage tools which are powered so that some of their movements are in direction other than along the line of travel
3. **multi powered tillage tools** – tillage tools powered by more than one form of power, such as draft and rotating power, or draft and electrical power
4. **simple tillage tools** – tillage tools which present a reasonable constant boundary area to the soil



NOMENCLATURE FOR TILLAGE TOOLS AND IMPLEMENTS

1. **side angle** - angle, in the soil surface plane, between a tool axis and a line, which is perpendicular to the direction of travel
2. **tilt angle** - angle, in a vertical plane perpendicular to the direction of travel, between a tool axis and the soil surface
3. **specific draft/unit draft** - draft force of an implement per unit area of tilled cross section
4. **soil tool geometry** - configuration of the soil tool boundary wherein the overall shape is usually oriented with the direction of travel of the tool and the soil surface
5. **soil additive applicator** - machine used to apply, or to apply and incorporate soil additives by means of tillage
6. **soil additive incorporator** - machine used to mechanically incorporate or mix material into the soil
7. **soil opener** - tillage tool used to slice through soil and create an opening for the insertion of material such as seeds, pesticides, fertilizers (disc, knife, and runner)
8. **soil roller** - rotating implement which pulverizes, firms or smooths soil by crushing or compacting



Height of soil above a
specified (initial or
subsequent) soil
surface

- 
- a. Ridge height
 - b. Bed height
 - c. Windrow height
 - d. All of the above



Height of soil above a
specified (initial or
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- a. Ridge height
 - b. Bed height
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A farmer plans to plant corn seeds in rows with a spacing of 40 cm between each row. The goal is to ensure proper alignment for future machinery operations, such as spraying or cultivating. Which type of planter should the farmer use?

- a. Broadcaster
 - b. Row-crop drill planter
 - c. Solid drill planter
 - d. Hill-drop planter
- 



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A farmer needs to cover a large area with grass seeds, but alignment in rows is not a priority. Instead, the seeds should be spread uniformly across the entire surface for full coverage. Which planting method is most suitable for this task?

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 - c. Solid drill planter
 - d. Hill-drop planter



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- 
- a. **Broadcaster**
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 - c. Solid drill planter
 - d. Hill-drop planter



PLANTING EQUIPMENT

- **Row-crop planters** – planters designed to plant seeds or plants in rows far enough apart to permit the subsequent entry of machinery.
 1. **Row-crop drill planter** – designed to plant seeds continuously in rows with row spacing greater than 36 cm.
 2. **Hill-drop planter** – designed to plant one or more seeds or plants in rows and in hills.
- **Solid planters** – planters designed to plant seeds or plants in patterns which do not allow subsequent entry of machinery.
 1. **Solid drill planter** – designed to plant seeds continuously in rows with row spacing less than 36 cm.
 2. **Broadcaster** – designed to scatter the seeds uniformly over the surface of the field without definite rows and hills. Seeds are covered by harrowing



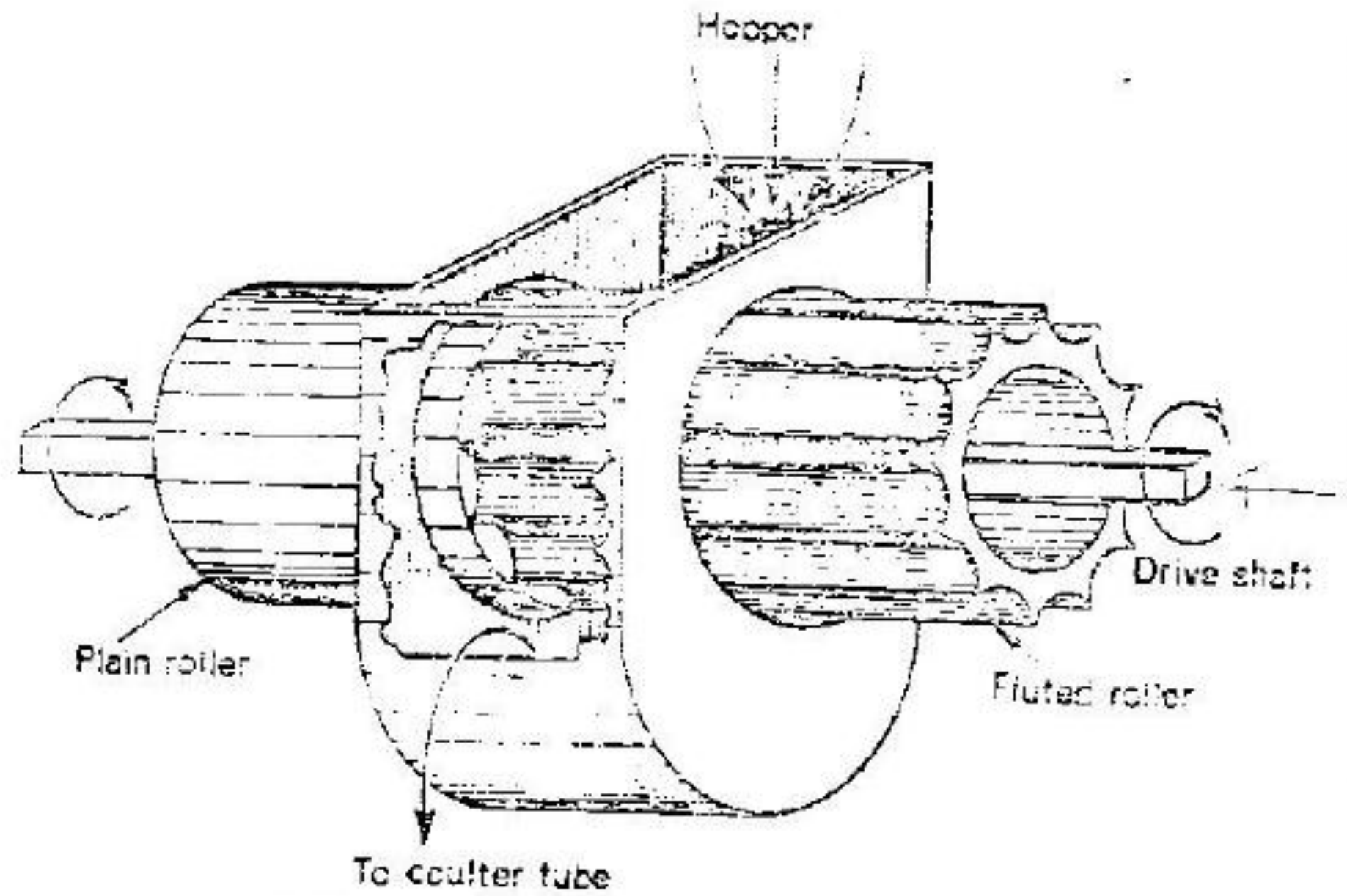
The ground wheel of a planter is normally related to the operation of the _____.

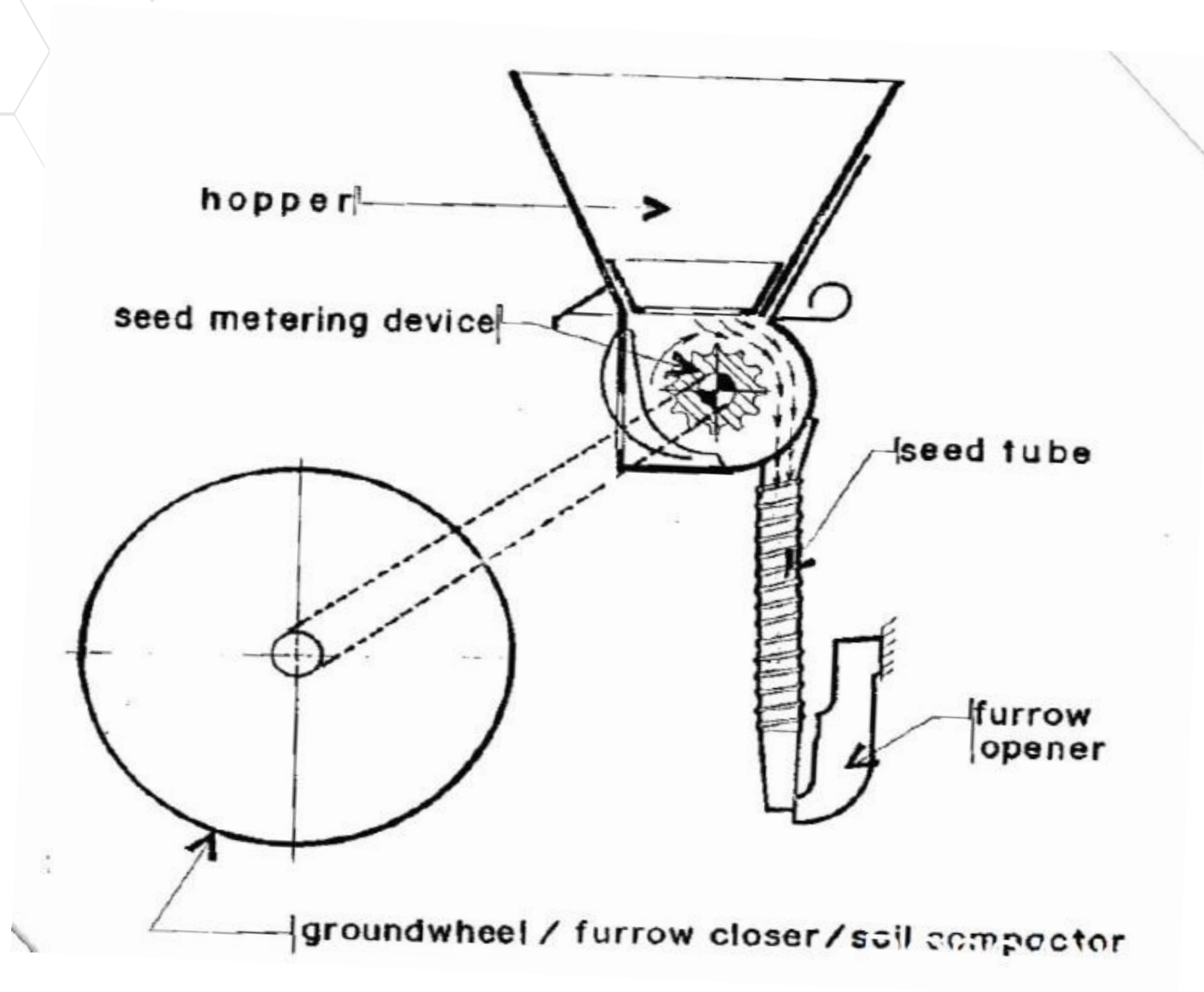
- a. Furrow opener
- b. Metering device
- c. Seed tube
- d. Double-disc closer



The ground wheel of a planter is normally related to the operation of the _____.

- a. Furrow opener
- b. Metering device**
- c. Seed tube
- d. Double-disc closer





Mechanical functions of a planter:

1. Make a furrow – **Furrow opener**
2. Meter the seeds – **Metering device**
3. Deposit the seeds – **Seed tube**
4. Cover the seeds – **Furrow closer**

The fertility of a field is such that maximum corn yields are obtained with a population of 72,000 plants per hectare. The rows are 0.85 m apart and an average emergence of 85 percent is expected. How many seeds per hill should be planted if the hills are 0.7 m apart?

- a. 4-5
- b. 5-6
- c. 6-7
- d. 7-8



CALIBRATION OF PLANTERS

$$\text{Expected Plant Population (EPP)} = \frac{\left(\frac{10,000 \text{ m}^2}{1 \text{ ha}}\right)(\text{Seeds per hill})(\text{Emergence})}{\text{Row Spacing} \times \text{Hill Spacing}}$$

CALIBRATION OF PLANTERS (Given: Groundwheel)

$$EPP = \frac{\left(\frac{10,000 \text{ m}^2}{1 \text{ ha}}\right)(\text{Seeds per hill})(\text{seed plate})(\text{Emergence})}{HS \times RS}$$

$$HS = \pi D \times SR$$



CALIBRATION OF PLANTERS (Given: Teeth)



$$\text{Expected Plant Population (EPP)} = \frac{\left(\frac{10,000 \text{ m}^2}{1 \text{ ha}}\right)(\text{Seeds per hill})(\text{Emergence})}{\text{RS} \times \text{HS}}$$

$$\text{HS} = \frac{\text{Circumference of groundwheel} \times \left(\frac{T_4}{T_3} \times \frac{T_2}{T_1}\right)}{\text{Seed Plate}}$$

The fertility of a field is such that maximum corn yields are obtained with a population of 72,000 plants per hectare. The rows are 0.85 m apart and an average emergence of 85 percent is expected. How many seeds per hill should be planted if the hills are 0.7 m apart?

$$\text{Expected Plant Population (EPP)} = \frac{\left(\frac{10,000 \text{ m}^2}{1 \text{ ha}}\right)(\text{Seeds per hill})(\text{Emergence})}{\text{Area per hill}}$$

$$72,000 \frac{\text{plants}}{\text{ha}} = \frac{\left(\frac{10,000 \text{ m}^2}{1 \text{ ha}}\right)(\text{Seeds per hill})(0.85)}{0.85 \text{ m} \times 0.7 \text{ m}}$$

$$\text{Seeds per hill} = 5.04 = 5 - 6 \text{ seeds per hill}$$

The fertility of a field is such that maximum corn yields are obtained with a population of 72,000 plants per hectare. The rows are 0.85 m apart and an average emergence of 75 percent is expected. How many seeds per hill should be planted if the hills are 0.7 m apart?

- a. 4-5
- b. 5-6**
- c. 6-7
- d. 7-8

If the speed ratio of the groundwheel over seedplate of a two-row corn planter is 5.0, determine the plant population per hectare. Row spacing is 75 cm. Groundwheel diameter is 60 cm. Ave. no. of seeds per hill is 2. No. of cells of seedplate is 25. Ave. emergence of seeds is 85%.

- a. 57,143.20 plants/ha
- b. 59,250 plants/ha
- c. 60,125.20 plants/ha
- d. 60,768.35 plants/ha

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$$EPP = \frac{\left(\frac{10,000 \text{ m}^2}{1 \text{ ha}}\right)(\text{Seeds per hill})(\text{seed plate})(\text{Emergence})}{(\pi D \times SR) \times RS}$$

$$EPP = \frac{\left(\frac{10,000 \text{ m}^2}{1 \text{ ha}}\right)(2)(25)(0.85)}{(\pi(0.6 \text{ m}) \times 5) \times 0.75 \text{ m}}$$

$$EPP = 60,125.20 \text{ plants/ha}$$



Using the specifications of the transmission system of the metering device (horizontal seed plate) of the two-row corn planter, determine the expected hill spacing. Row spacing = 75 cm, groundwheel diameter = 60 cm, no. of cells of seedplate = 20, $T1 = 6t$, $T2 = 18t$, $T3 = 10t$ and $T4 = 25t$

- a. 0.62 m
- b. 0.81 m
- c. 0.71m
- d. 0.92 m

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$$HS = \frac{\text{Circumference of groundwheel} \times \left(\frac{T_4}{T_3} \times \frac{T_2}{T_1}\right)}{\text{Seed Plate}}$$

$$HS = \frac{\pi(0.6 \text{ m}) \times \left(\frac{25}{10} \times \frac{18}{6}\right)}{20}$$

$$HS = 0.7069 \text{ m or } 0.71 \text{ m}$$



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- a. 0.62 m
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- d. 0.92 m

Using the results of the calibration test of the 9 X 6 inches grain drill, determine the seeding rate adjustment to use at $\frac{3}{4}$ setting if it is desired to plant at the rate of 100 kg per hectare. Groundwheel diameter of 1.22 m.

ADJUSTMENT	DISCHARGE/10 REV (GPM)
0	0
$\frac{1}{4}$	120
$\frac{1}{2}$	470
$\frac{3}{4}$	810
Full	1100

- a. 22.83 kg/ha
- b. 89.4 kg/ha
- c. 154.08 kg/ha
- d. 209.24 kg/ha



CALIBRATION OF PLANTERS (Seeding Adjustment)

Area (covered by ground wheel) = Width x Distance

Width = Rows x Row Length

Distance = πD x No. of rev

$$\text{Adjusted seed}_{\text{setting}} = \text{Discharge}_{\text{setting}} \times \frac{10,000 \text{ m}^2}{\text{Area}}$$

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ADJUSTMENT	DISCHARGE/10 REV (GPM)
0	0
$\frac{1}{4}$	120
$\frac{1}{2}$	470
$\frac{3}{4}$	810
Full	1100

Width = Rows x Row Length

$$\text{Width} = 9 \text{ rows} \times 6 \text{ inches} \left(\frac{0.0254 \text{ m}}{1 \text{ inch}} \right) = 1.3716 \text{ m}$$

Distance = $\pi D \times \text{No. of rev}$

$$\text{Distance} = \pi(1.22 \text{ m}) \times 10 = \frac{61}{5} \pi \text{ m}$$

$$\text{Area} = \text{Width} \times \text{Distance} = 1.3716 \text{ m} \times \frac{61}{5} \pi \text{ m} = 52.57 \text{ m}^2$$

$$\text{Adjusted seed}_{3/4} = \text{Discharge}_{3/4} \times \frac{10,000 \text{ m}^2}{\text{Area}}$$

$$\text{Adjusted seed}_{3/4} = 0.81 \text{ kg} \times \frac{10,000 \text{ m}^2/\text{hs}}{52.57 \text{ m}^2} = \mathbf{154.08 \text{ kg/ha}}$$

Using the results of the calibration test of the 9 X 6 inches grain drill, determine the seeding rate adjustment to use at $\frac{3}{4}$ setting if it is desired to plant at the rate of 100 kg per hectare. Groundwheel diameter of 1.22 m.

ADJUSTMENT	DISCHARGE/10 REV (GPM)
0	0
$\frac{1}{4}$	120
$\frac{1}{2}$	470
$\frac{3}{4}$	810
Full	1100

- a. 22.83 kg/ha
- b. 89.4 kg/ha
- c. 154.08 kg/ha
- d. 209.24 kg/ha

A field measuring 45x90m is to be prepared using a 1m upland power tiller at a speed of 3kph. Assume a 10% overlap between passing's and 9 seconds turnaround time at the end of each pass. Determine actual field capacity when tilling is done along the longer field direction.

- a. 0.25 ha/hr
- b. 0.5 ha/hr
- c. 0.7 ha/hr
- d. 1 ha/hr

A field measuring 45x90m is to be prepared using a 1m upland power tiller at a speed of 3kph. Assume a 10% overlap between passing's and 9 seconds turnaround time at the end of each pass. Determine actual field capacity when tilling is done along the longer field direction.

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$$\text{Number of rows} = 45 \text{ m} / (1 \text{ m} \times 0.9) = 50 \text{ rows}$$

$$\text{Total Area} = 45 \text{ m} \times 90 \text{ m} = 0.405 \text{ ha}$$

$$T1 \text{ (Time to turn 50 rows)} = 50 \text{ rows} (9 \text{ sec/row}) = 450 \text{ sec or } 0.125 \text{ hr}$$

$$T2 \text{ (Time to cover length)} = (90 \text{ m} \times 50 \text{ rows}) / (3000 \text{ m/hr}) = 1.5 \text{ hr}$$

$$\text{Total Time} = T1 + T2 = 1.625 \text{ hr}$$

$$\text{AFC} = (0.405 \text{ ha}) / 1.625 \text{ hr} = \mathbf{0.25 \text{ ha/hr}}$$



A 1.2m wide floating tiller is used for rotary tilling/pudding a 0.1008 ha wetland paddy field. The average speed is 3kph and turning at the headlands takes 10 seconds. What is the field efficiency if the field is worked along its length?

- a. 66.67%
- b. 71.64%
- c. 83.33%
- d. 88.67%

A 1.2m wide floating tiller is used for rotary tilling/pudding a 0.1008 ha wetland paddy field. The average speed is 3kph and turning at the headlands takes 10 seconds. What is the field efficiency if the field is worked along its length?

Note: No dimensions are given. Only the area of field. The only way to find the dimensions is by trial and error.

$$0.1008 \text{ ha} = 1008 \text{ sq. m}$$

$$\text{Let: } L = 2W$$

$$A = L \times W$$

$$A = 2W \times W$$

$$1008 = 2W \times W$$

$$W = 22.4 \text{ m}$$

$$\text{Thus, } W = 22.4 \text{ m and } L = 44.9 \text{ m}$$

You can use between L and W as a baseline. The goal is to find the length that will yield a whole number dimension when it is divided to the area:

I'll use $L = 44.9$ as a baseline.

We go higher:

$$1008/45 = 22.4 \text{ m}$$

$$1008/46 = 21.9 \text{ m}$$

$$1008/47 = 21.4 \text{ m}$$

We go lower:

$$1008/44 = 22.9 \text{ m}$$

$$1008/43 = 23.4 \text{ m}$$

$$1008/42 = 24 \text{ m}$$

So, our dimension now is 24 m x 42 m

A 1.2m wide floating tiller is used for rotary tilling/pudding a 0.1008 ha wetland paddy field. The average speed is 3kph and turning at the headlands takes 10 seconds. What is the field efficiency if the field is worked along its length?

$$\text{Number of rows} = 24 \text{ m} / (1.2 \text{ m}) = 20 \text{ rows}$$

$$T1 \text{ (Time to turn 20 rows)} = 20 \text{ rows} (10 \text{ sec/row}) = 200 \text{ sec or } 0.056 \text{ hr}$$

$$T2 \text{ (Time to cover length)} = (42 \text{ m} \times 20 \text{ rows}) / (3000 \text{ m/hr}) = 0.28 \text{ hr}$$

$$\text{Total Time} = T1 + T2 = 0.336 \text{ hr}$$

$$\text{AFC} = (0.1008 \text{ ha/hr}) / 0.336 \text{ hr} = 0.3 \text{ ha/hr}$$

$$\text{TFC} = \text{SWEff} / 10 = (3 \text{ kph})(1.2) / 10 = 0.36 \text{ ha/hr}$$

$$\text{Field Eff} = \text{AFC} / \text{TFC} = 0.3 / 0.36 = \mathbf{83.33\%}$$



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During which stage of crop growth is the plant particularly vulnerable to the effects of weeds, pests, and diseases, especially impacting yield?

- 
- a. Seedling stage
 - b. Vegetative stage
 - c. Flowering stage
 - d. Maturity stage



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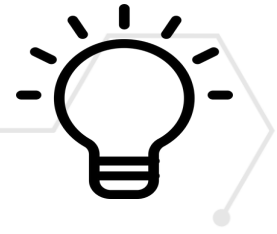
A farmer is facing a significant problem with a particular pest that is damaging his corn crop. He decides to use a method that involves introducing a specific insect that naturally preys on this pest to help manage the infestation without harming the corn plants. Which pest management strategy is the farmer employing?

- a. Cultural control
- b. Biological control
- c. Chemical control
- d. Mechanical control



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- a. Cultural control
 - b. Biological control**
 - c. Chemical control
 - d. Mechanical control



Crop protection – process of providing plants the conditions that will make them free of weeds, pests and diseases.

METHODS OF PEST CONTROL

1. **Cultural control** – Adjusting practices like planting time and crop rotation.
2. **Ecological control** – Modifying the environment (e.g., flooding to kill pests).
3. **Biological control** – Using insects, or repellent plants to manage pests.
4. **Physiological control** – Breeding and planting pest-resistant varieties.
5. **Chemical control** – Applying chemicals to manage weeds, pests, and diseases.
6. **Flame control** – Using fire to selectively burn weeds without harming crops.
7. **Mechanical control** – Using tools and machines for land prep, weeding, and pest control.



Which of the following is **NOT** a function of a sprayer?

- a. Break the liquid into droplets of effective size
- b. Distribute the spray solution uniformly
- c. Regulate the spraying rate
- d. Mix the spray solution with the soil during application



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- a. Break the liquid into droplets of effective size
- b. Distribute the spray solution uniformly
- c. Regulate the spraying rate
- d. Mix the spray solution with the soil during application



SPRAYER



- Most popular way of applying chemicals in the Philippines is with the use of knapsack sprayers.
- Sprayers are used primarily for the application of herbicides, insecticides, fungicides and plant nutrients.

Functions of a sprayer

1. Break the liquid into droplets of effective size
2. Distribute the spray solution uniformly
3. Regulate the spraying rate



In a sprayer, which component serves as a rigid conductor that delivers the spray from the hose to the nozzle?

- a. Lance
- b. Boom
- c. Wand
- d. Pipe handle

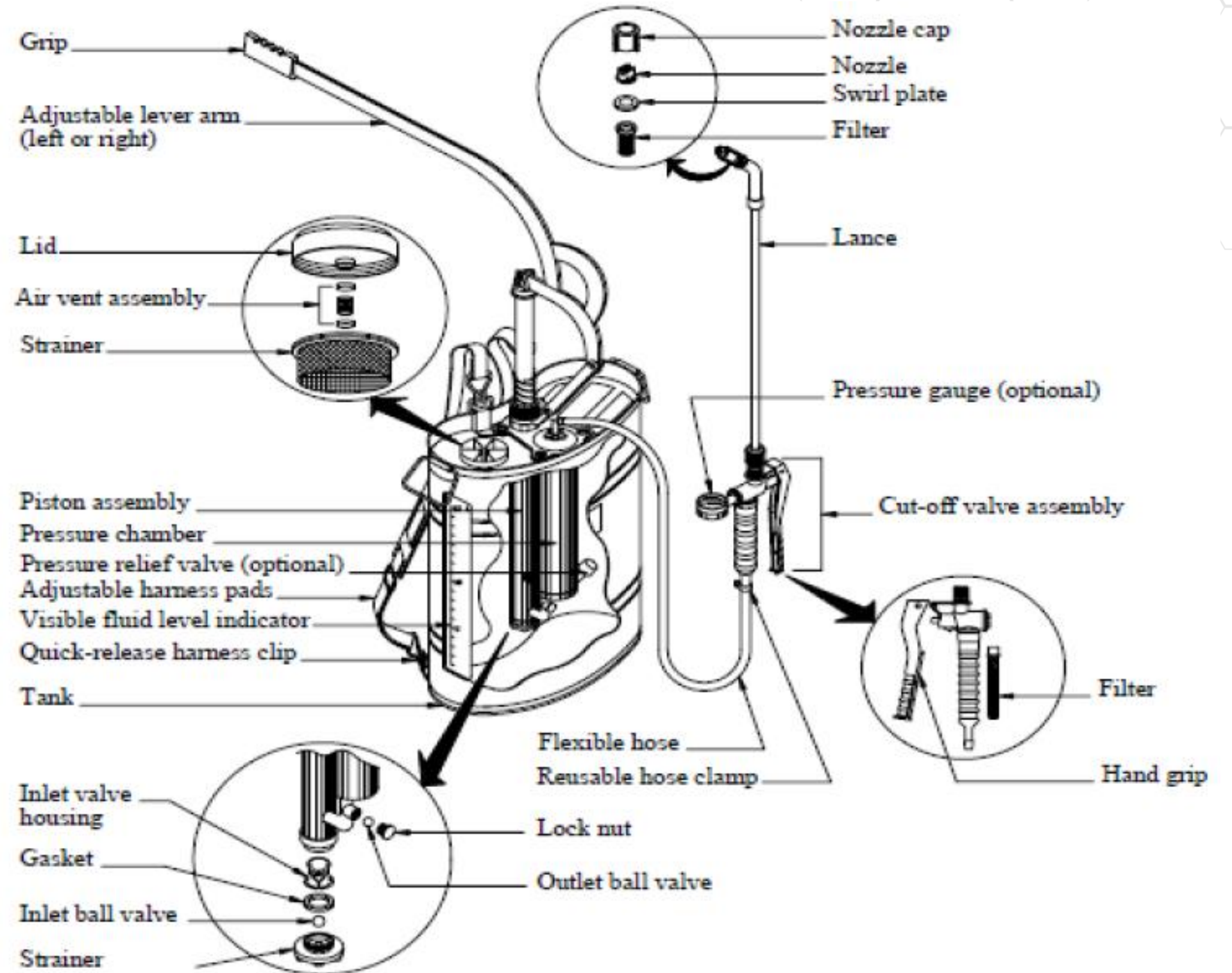


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- c. Wand
- d. Pipe handle

Basic Components of Sprayer

1. Chemical supply : Tank
2. Energy source : Pump and pressure chamber
3. Atomizer : Nozzles
4. Control device : Cut-off valve
5. Conductors : Flexible hose and rigid lance





In which type of sprayer is pressure created by adding air to a container with a fixed volume of liquid?

- a. Knapsack sprayer
- b. Compression sprayer
- c. Pneumatic sprayer
- d. Hydraulic sprayer



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- d. Hydraulic sprayer



Principles of Creating Hydraulic Energy



1. Liquid → Air Space

- ✓ Add liquid into a chamber with fixed air volume
- ✓ *Example: Knapsack sprayer*

2. Air → Liquid Container

- ✓ Add air into a container with fixed liquid volume
- ✓ *Example: Compression sprayer*

A farmer notices that the pesticide application on his field is inconsistent, resulting in some areas being over-treated and others under-treated. Upon reviewing his spraying process, he identifies three key factors:

1. Nozzle discharge at different pumping rates (L/min)
2. Effective working width or swath (m)
3. Walking speed (m/min)

- a. Nozzle discharge at different pumping rates
- b. Effective working width or swath
- c. Walking speed
- d. Weather conditions

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A farmer is applying herbicide to a flat surface area. Which nozzle type is most suitable for this task?

- a. Cone type nozzle
- b. Fan type nozzle
- c. Hollow cone nozzle
- d. Solid stream nozzle



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- d. Solid stream nozzle

Nozzles

Performance is dependent on hydraulic energy as follows:

1. The higher the pressure, the smaller the droplets
2. The higher the pressure, the higher the flow rate
3. The higher the pressure, the wider the spray angle

Types of nozzles

1. Cone type – produces a cone shaped pattern of spray which could be hollow or solid cone.

- Best suited for spraying crops because it produces a spray in which droplets approach the leaves from several angles.
- The pressure is high and then orifice is small.

2. Fan type – produces a flat pattern.

- Best suited for spraying flat surfaces as in the application of herbicides.
- The pressure is low and the orifice is large.



This spraying method is usually used for aviation spraying.

- a. Mist
- b. Ultra-low-volume
- c. Powder
- d. Smoke



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- a. Mist
- b. Ultra-low-volume**
- c. Powder
- d. Smoke

APPLICATION METHODS

Mist spraying method

- Called fog sprayer (knapsack)
- use the high speed air to blow out the liquid, then impact on the air to form small droplets attach to the crops with 50–100 μm in diameter

Ultralow-volume spraying method

- Use high concentration of pesticide, by the centrifugal force the liquid are scattered into tiny droplets with 70–300 μm in diameter
- Usually method used for aviation spraying

Powder spraying method

- Use high speed of airflow to blow the powder to the crop

Smoke spraying method

- Refers to high-speed rotating air to form liquid to 20 μm droplets, then blow out.
- Used for some enclosed space such as Greenhouse

A farmer wishes to spray his vegetable farm at a rate of 300 gallons per hectare. 16 nozzles, 18 inches apart will be used. The forward speed of the tractor is at 5 kph.

Compute the field capacity of the sprayer in ha/hr.

- a. 2.4 ha/hr b. 3.6 ha/hr c. 4.8 ha/hr d. 5.1 ha/hr

What is the required pump capacity for the sprayer.

- a. 16 gpm b. 17 gpm c. 18 gpm d. 19 gpm

What is the capacity per nozzle?

- a. 0.67 gpm b. 1.3 gpm c. 1.8 gpm d. 2 gpm

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Compute the field capacity of the sprayer in ha/hr.

$$FC = \frac{SWEff}{10}$$

$$FC = \frac{(5 \text{ kph})(16 \times 18 \text{ in} \times \frac{0.0254 \text{ m}}{1 \text{ in}})}{10}$$

$$FC = 3.6576 \text{ ha/hr}$$

$$\text{Field Capacity} = \frac{SWEff}{10}$$

S (speed) = kph
W (Width) = m
Eff (Efficiency) = decimal

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What is the required pump capacity for the sprayer.

For sprayer:

Pump Capacity (Q) = Application rate x Swath x Speed

Pump Capacity (Q) = Application rate x FC

$$Q = \text{Application Rate} \times \text{FC}$$

$$Q = \frac{300 \text{ gal}}{\text{hr}} \times \frac{3.6 \text{ ha}}{\text{hr}} \times \frac{1 \text{ hr}}{60 \text{ min}}$$

$$Q = 18 \text{ gpm}$$

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What is the capacity per nozzle?

$$Q_n = Q / \text{no. of nozzle}$$

$$Q_n = 18 \text{ gpm} / 16 \text{ nozzle}$$

$$Q_n = 1.125 \text{ gpm/nozzle}$$

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Walking in the rice field operating the sprayer with a spray swath of 1 m, a person covers 20 m/min. If nozzle discharge is 0.4 L/min, what is the application rate?

- a. 0.2 L/m^2
- b. 200 L/ha
- c. $20 \text{ m}^3/\text{ha}$
- d. 2 L/ha

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For sprayer: $Q = \text{Application rate} \times \text{Swath} \times \text{Speed}$

$0.4 \text{ L/min} = \text{Application rate} \times 1 \text{ m} \times 20 \text{ m/min}$

Application rate = 0.02 L/m^2 or 200 L/ha

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- a. 0.2 L/m^2
- b. 200 L/ha
- c. $20 \text{ m}^3/\text{ha}$
- d. 2 L/ha



VOUME OF WATER APPLIED:

$$\text{Vol. of water} = \frac{\frac{10,000 \text{ m}^2}{1 \text{ ha}} (\text{Nozzle Discharge})}{\text{Swath} \times \text{Speed}}$$

AMOUNT OF CHEMICAL TO MIX WITH WATER:

$$\text{Amount of chemical} = \frac{\text{Swath} \times \text{Speed} \times \text{Capacity of Tank} \times \text{Chemical app. rate}}{q \times \frac{10,000 \text{ m}^2}{1 \text{ ha}}}$$

The application rate of a certain chemical is recommended at 4 kg/ha. The calibration test of a 16-liter knapsack sprayer is as follows: Nozzle discharge = 1.2 liters/min, swath = 1.5 m and walking speed = 20 m/min Determine the total volume of water applied per hectare.

- a. 320 L/ha
- b. 360 L/ha
- c. 400 L/ha
- d. 440 L/ha

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$$\text{Vol. of water} = \frac{\frac{10,000 \text{ m}^2}{1 \text{ ha}} (1.2 \text{ L/min})}{1.5 \text{ m} \times 20 \text{ m/min}}$$

$$\text{Vol. of water} = 400 \text{ L/ha}$$

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- a. 102 gms/load
- b. 160 gms/load
- c. 200 gms/load
- d. 240 gms/load

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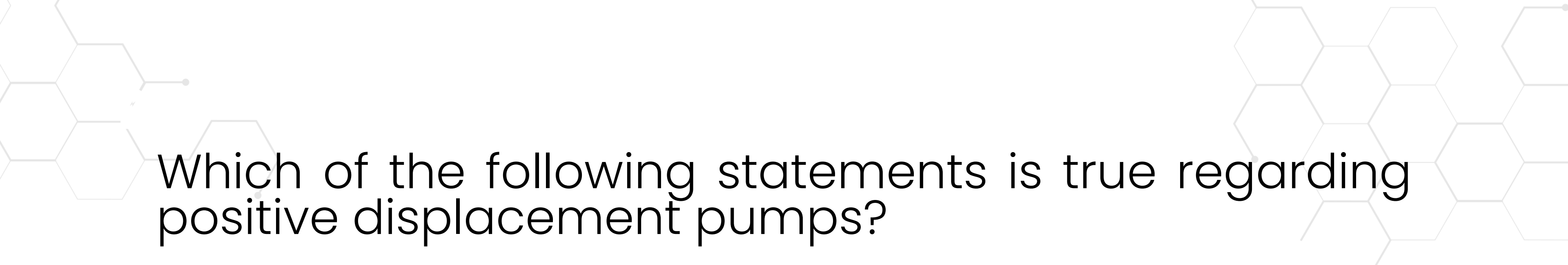
$$\text{Amount of chemical} = \frac{\text{Swath} \times \text{Speed} \times \text{Capacity of Tank} \times \text{Chemical App. rate}}{q \times \frac{10,000 \text{ m}^2}{1 \text{ ha}}}$$

$$\text{Amount of chemical} = \frac{1.5 \text{ m} \times 20 \text{ m/min} \times 16 \text{ L} \times 4 \text{ kg/ha}}{1.2 \text{ L/min} \times \frac{10,000 \text{ m}^2}{1 \text{ ha}}}$$

$$\text{Amount of chemical} = 0.16 \text{ kg or 160 grams}$$

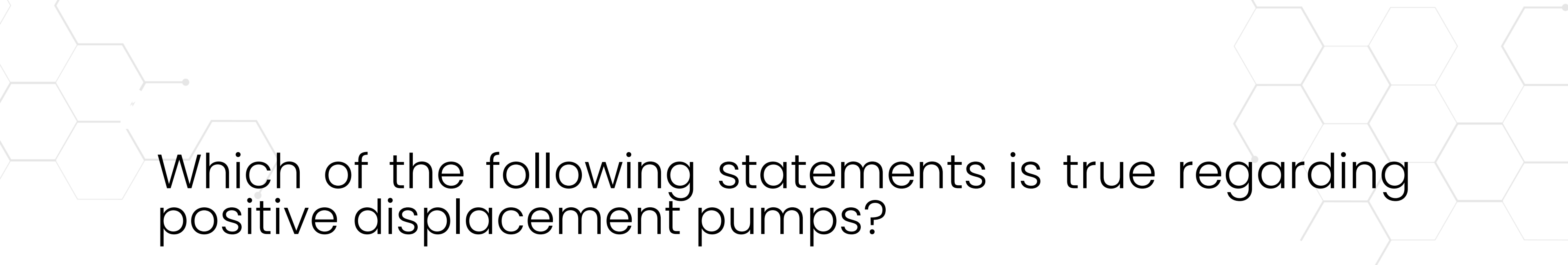
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Which of the following statements is true regarding positive displacement pumps?

- a. They have an inverse relationship between discharge rate and pressure head.
- b. It discharge the same volume of water regardless of the head against which they operate.
- c. These are primarily used for high-flow applications.
- d. These are less efficient than centrifugal pumps at all operating conditions.



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Water Pumping Equipment



Pumps

- Device for transferring fluids from lower to higher elevation
- Common water sources: wells, rivers, lakes, reservoirs, canals
- Essential for health, agriculture, and irrigation
- Can be **hand-operated** (domestic) or **power-operated** (domestic & irrigation)

Pump Classifications

A. Positive Displacement Pumps

- Discharge **same volume** regardless of head pressure
- **Reciprocating Pump** – to-and-fro motion
- **Basic parts:** piston/plunger, inlet valve, outlet valve

Types:

- **Lift Pump** – single-acting, lifts water to spout
- **Force Pump** – single/double-acting, forces water above atmospheric pressure

B. Variable Displacement Pumps

- Inverse relation: higher head → lower discharge
 - **Centrifugal Pump** – common for domestic & irrigation; uses centrifugal force; impeller in casing
 - **Propeller Pump** – low head, high discharge; for drainage & canal/river water transfer



What is a widely accepted criterion for determining the required pump capacity in agricultural irrigation?

- a. The average monthly water requirement of the crop
- b. The lowest daily water requirement of the crop
- c. The highest daily water requirement of the crop
- d. The total seasonal water requirement of the crop



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What are the two parameters used in selecting size of pumps?

- a. Discharge and power requirement
- b. Head and mass flow rate
- c. Pressure and area of pipe
- d. Head and discharge



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What is the theoretical maximum height that water can be lifted using suction?

- a. 2.31 ft
- b. 33 ft
- c. 3.3 m
- d. None of the above



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- a. 2.31 ft
- b. 33 ft**
- c. 3.3 m
- d. None of the above

Basic Pump Operating Characteristics

1. **Discharge** – Volume of water delivered per unit of time
2. **Head** – Height of a vertical column of water; Represents energy content of water per unit weight, referenced to an arbitrary datum
3. **Total Static Head** – Total vertical distance water is lifted; Can be measured directly from water surface to discharge point
4. **Friction Head** – Energy loss due to friction in pipes and fittings. Includes friction from: Pipes, elbows, valves, etc.
5. **Velocity Head** – Energy due to water's velocity; Usually small and negligible in irrigation loss computations
6. **Suction Head** – Pump above water surface → works with suction head; Suction head includes: Vertical suction lift & friction losses in suction side components
7. **Net Positive Suction Head (NPSH)** – sets the allowable suction limit
 - ❑ Theoretical max suction lift: **33 feet**



Pump and driver share common shaft & bearings and forms a single compact unit

- a. Close-coupled drive
- b. Belt-driven drive
- c. Common shaft drive
- d. Two of the choices above



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COUPLING

Coupling

- ✓ Connects a **prime mover** to driven machinery
- ✓ Transmit **rotary motion** and **torque**
 - ❖ Accommodate misalignment
 - ❖ Absorb shock loads
 - ❖ Provide flexibility between connected equipment

Types of Pump Drives

1. **Close-Coupled Drive** – Pump and driver share **common shaft & bearings**; forms a **single compact unit**
2. **Belt-Driven Drive** – Pump equipped with **pulley head** for belt drive

A decorative background element consisting of a light gray hexagonal pattern, resembling a honeycomb or molecular structure, located at the top of the slide.

1000 gpm of water is pumped from a river to a farm 150 ft above river. What is required HP to drive pump, considering a pump efficiency of 70% and friction and other losses of 20% of static head.

- a. 57.1 hp
- b. 64.9 hp
- c. 72.3 hp
- d. 81.6 hp

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$$H = 150 \text{ ft} + 150 \text{ ft}(0.20) = 180 \text{ ft}$$

$$P = \frac{QH}{3960 \times \text{Eff}}$$
$$P = \frac{(1000 \text{ gpm})(180 \text{ ft})}{3960 \times 0.7}$$

$$\text{HP} = 64.9 \text{ hp}$$

$$P = \frac{QH}{3960 \times \text{Eff}}$$

Q (discharge) = gpm

H (head) = ft

Eff (Efficiency) = decimal

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- d. 81.6 hp



A pump discharges 4 lps at a head of 6 m is driven by an electric motor. The input current of motor is 1.5 amp while input voltage is 220 V. What is the overall efficiency of pump if motor power factor is 0.98.

- a. 56.32%
- b. 61.09%
- c. 72.7%
- d. 76.3%



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- c. 72.7%
- d. 76.3%

A pump discharges 4 lps at a head of 6 m is driven by an electric motor. The input current of motor is 1.5 amp while input voltage is 220 V. What is the overall efficiency of pump if motor power factor is 0.98.

$$P = \frac{QH}{3960 \times \text{Eff}}$$
$$P = \frac{\left(\frac{4\text{L}}{\text{sec}} \times \frac{1\text{ gal}}{3.785\text{ L}} \times \frac{60\text{ sec}}{1\text{ min}}\right)(6\text{ m} \times \frac{3.28\text{ ft}}{1\text{ m}})}{3960} = 0.3151\text{ hp or }235.07\text{ W}$$

$$P = IV \times \text{power factor}$$

$$P = 220\text{ V} \times 1.5\text{ A} \times 0.98 = 323.4\text{ W}$$

$$\text{Eff} = \frac{\text{Output}}{\text{Input}} = \frac{235.07\text{ W}}{323.4\text{ W}} = \mathbf{0.727 \text{ or } 72.7\%}$$

Determine the size of the centrifugal pump and air-cooled diesel engine to use for a 1.5 ha field planted to lowland rice. The field is to be irrigated for 12 hours every 2 days. Source of water is a well with the following data:

- a. Static water level = 8 ft
- b. Specific yield = 12 gpm/ft DD
- c. Discharge head = 0 ft
- d. Friction and velocity head = 2 ft
- e. Efficiency of transmission, pump, air-cooled diesel = 0.95, 0.70, 0.52
- f. Discharge = 120 gpm

a. 1HP b. 1.75 HP c. 2HP d. 2.5 HP


$$P = \frac{QH}{3960 \times \text{Eff}}$$

$$Q = 120 \text{ gpm}$$

H = Total head = Static head + drawdown + friction/velocity head + discharge head

$$H = 8 \text{ ft} + (120 \text{ gpm}/12 \text{ gpm/ft}) + 2 \text{ ft} + 0 \text{ ft} = 20 \text{ ft}$$

$$P = \frac{(120 \text{ gpm} \times 20 \text{ ft})}{3960 \times 0.52 \times 0.95 \times 0.70} = 1.75 \text{ HP}$$

Since power units are available in standard sizes only, select the next higher standard size of air cooled diesel engine. And that is **2 HP**.

Determine the size of the centrifugal pump and air-cooled diesel engine to use for a 1.5 ha field planted to lowland rice. The field is to be irrigated for 12 hours every 2 days. Source of water is a well with the following data:

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a. 1HP b. 1.75 HP c. 2HP d. 2.5 HP



According to IRRI, how many liters of water are required to produce 1 kilogram of rice in an irrigated lowland production system?

- a. 1,324 liters
 - b. 1,432 liters
 - c. 1,423 liters
 - d. 1,342 liters
- 



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- a. 1,324 liters
 - b. 1,432 liters**
 - c. 1,423 liters
 - d. 1,342 liters
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On average, it takes **1,432 liters** of water to produce 1 kg of rice in an irrigated lowland production system.

A general rule is that a rice crop will need **approximately 10 mm** of water per day. Therefore, a crop that matures in 100 days will require approximately **1000mm** of water while a crop that matures in 150 days will require 50% more

Source: IRRI



Which type of harvester cuts the standing crop and lays it in an orderly manner without binding it?

- 
- a. Hand tools
 - b. Reaper-windrower
 - c. Reaper-binder
 - d. Combine harvester



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What cutting mechanism involves a scissor-type action between moving and stationary blades?

- 
- a. Slicing/tearing action
 - b. Single-element impact action
 - c. Two-element scissor-type action
 - d. High-speed rotary slicing



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Which of the following shows the correct sequence of operations in a combine stripper?

- a. Cutting → Separating rows → Conveying → Threshing → Shaking straw → Cleaning grains → Conveying grains
- b. Separating rows → Cutting → Conveying → Threshing → Shaking straw → Cleaning grains → Conveying grains
- c. Cutting → Threshing → Separating rows → Conveying → Shaking straw → Cleaning grains → Conveying grains
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HARVESTING AND THRESHING EQUIPMENT



- **Harvesting** – process of gathering the useful portion of the crop from the field
- **Threshing** – process of separating the grains from the straw. Additionally, it may include cleaning the grains.

Combine:

1. Separate rows
2. Cut standing crop
3. Convey cut material to threshing section
4. Threshing
5. Shake straw to separate loose grains
6. Clean grains of chaff
7. Convey grains to tank/sacks

Stripper:

1. Separate rows
2. Strip grains from plants
3. Clean grains
4. Convey grains to tank/sacks

Groupings of harvesters

1. **Hand tools** – include the sickle, scythe, yatab, lingkao and cradle.
2. **Reapers-windrower** – a machine that cuts the standing crop, conveys the cut crop to one side, and lays them down in an orderly manner.
3. **Reaper-binder** – a machine that cuts the standing crop, binds the cut crop, and lays down the bound crop to one side.

Cutting mechanisms

1. **Slicing/tearing action** – employed in hand tools
2. **High velocity, single-element, impact action** – use of sharp or dull-edged blades moving at high velocity of 2,000 fpm to 9,000 fpm
3. **Two-element, scissor type action** – shearing action between the moving and stationary blades

Which of the following statements about determining the correct harvesting time is false?

- a. The ideal moisture content for harvesting is between 20 and 25 percent (wet basis).
- b. Grains should be harvested when 60 to 65 percent of the grains per panicle are straw-colored.
- c. Late-maturing varieties should be harvested between 130 to 136 days after sowing.
- d. For wet season harvesting, the ideal time is between 32 to 38 days after heading.

Which of the following statements about determining the correct harvesting time is false?

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- b. Grains should be harvested when 60 to 65 percent of the grains per panicle are straw-colored.
- c. Late-maturing varieties should be harvested between 130 to 136 days after sowing.
- d. For wet season harvesting, the ideal time is between 32 to 38 days after heading.

How to Determine the Correct Harvesting Time

1. **Moisture Content (MC):** 20–25% (wet basis); grains should be firm but not brittle.
2. **Ripe Grains per Panicle:** Harvest when 80–85% of grains turn yellow (straw-colored).
3. **Days After Sowing:**
 - Late varieties: 130–136 days
 - Mid-maturing: 113–125 days
 - Early-maturing: 110 days
4. **Days After Heading:**
 - Dry season: 28–35 days
 - Wet season: 32–38 days
5. **Other Considerations:** Harvest early to avoid rewetting and grain breakage. Wait for surface moisture to evaporate if crops are wet.

Source: Reference Manual on Harvesting & Threshing of PhilMech

Pre-Operation Procedures for Mechanical Reaping

1. **Plan the field pattern:** Determine entry and exit points.
2. **Drain the field:** 7 to 10 days before harvest or when grains are in the hard dough stage.
3. **Manual cutting:** Cut the area to the size of the header at the entry point.
4. **Trim edges:** Manually cut crops at corners and edges not reachable by the machine.

A 4 m wide combine travelling at 6 kph can empty its 2-ton tanks in 3 minutes. When unloading on-the-go, it has 80% field efficiency. Field yield is 4 tons/ha. What would be the efficiency if the combine stopped to unload.

- a. 66.67%
- b. 71.43%
- c. 78.10%
- d. 78.09%

A 4 m wide combine travelling at 6 kph can empty its 2-ton tanks in 3 minutes. When unloading on-the-go, it has 80% field efficiency. Field yield is 4 tons/ha. What would be the efficiency if the combine stopped to unload.

Unloading on-the-go:

$$FC \text{ (on-the-go)} = \frac{SWEff}{10} = \frac{(6 \text{ kph})(4m)(0.80)}{10} = 1.92 \text{ ha/hr}$$

For t_1 :

$$t_1 = \text{Eff} \times \text{operating time} = 0.8 \times 3600 \text{ sec} = 2880 \text{ sec}$$

For t_2 :

$$\text{No. of unloadings/hr: } \frac{4 \text{ tons}}{\text{ha}} \times \frac{1.92 \text{ ha}}{\text{hr}} \times \frac{1}{2 \text{ tons/load}} = 3.84 \text{ or } 4 \text{ unloadings/hr}$$

$$t_2 = 4 \text{ unloadings/hr} \times 180 \text{ sec/unloading} = 720 \text{ sec}$$

$$\text{New eff} = \frac{t_1}{T+t_2} = \frac{2880}{3600+720} = \mathbf{66.67\%}$$

A 4 m wide combine travelling at 6 kph can empty its 2-ton tanks in 3 minutes. When unloading on-the-go, it has 80% field efficiency. Field yield is 4 tons/ha. What would be the efficiency if the combine stopped to unload.

- a. 66.67%
- b. 71.43%
- c. 78.10%
- d. 78.09%

A 5-m self-propelled combine makes an average stop of 4 minutes every time its 2-ton grain tank is to be unloaded. The yield of the 20-ha field is 40 tons. The operating speed is 4.8 kph. The time for turning on the headland at the ends of the 500-m field is 15 seconds. What is the field efficiency?

- a. 87.67%
- b. 86.25%
- c. 85.19%
- d. 83.33%

A 6-m self-propelled combine makes an average stop of 5 minutes every time its 2-ton grain tank is to be unloaded. The yield of the 24-ha field is 30 tons. The operating speed is 4.0 kph. The time for turning on the headland at the ends of the 500-m field is 15 seconds. What is the field efficiency?

- a. 87.67%
- b. 86.25%
- c. 85.19%
- d. 83.33%

A 6-m self-propelled combine makes an average stop of 5 minutes every time its 2-ton grain tank is to be unloaded. The yield of the 24-ha field is 30 tons. The operating speed is 4 kph. The time for turning on the headland at the ends of the 500-m field is 15 seconds. What is the field efficiency?

For TFC:

$$\text{TFC} = \frac{\text{SWEff}}{10}$$

$$\text{TFC} = \frac{(4 \text{ kph})(6 \text{ m})(1)}{10}$$

$$\text{TFC} = 2.4 \frac{\text{ha}}{\text{hr}}$$

For AFC:

No. of rows = $480 \text{ m} / 6 \text{ m/row} = 80 \text{ rows}$

For t_1 : Actual working time in the rows at 100% efficiency

$$t_1 = \frac{\text{distance}}{\text{Rate}} = \frac{80 \text{ rows} \times \frac{500 \text{ m}}{\text{row}}}{\frac{4 \text{ km}}{\text{hr}} \times \frac{1000 \text{ m}}{1 \text{ km}}} = 10 \text{ hr}$$

For t_2 : time for turning at headland

$$t_1 = 80 \text{ rows} \times \frac{1 \text{ turn}}{\text{row}} \times \frac{15 \text{ sec}}{\text{turn}} \times \frac{1 \text{ hr}}{3600 \text{ sec}} = 0.33 \text{ hr}$$

For t_3 : time for unloading of tank

$$t_1 = 30 \text{ tons} \times \frac{1}{2 \text{ tons/tank}} \times \frac{5 \text{ min}}{\text{tank}} \times \frac{1 \text{ hr}}{60 \text{ min}} = 1.25 \text{ hr}$$


$$T = t_1 + t_2 + t_3$$

$$T = 10 \text{ hr} + 0.33 \text{ hr} + 1.25 \text{ hr} = 11.58 \text{ hr}$$

$$AFC = \frac{A}{T} = \frac{24 \text{ ha}}{11.58 \text{ hr}} = 2.07 \text{ ha/hr}$$

$$\text{Eff} = \frac{AFC}{TFC} = \frac{2.07 \text{ ha/hr}}{2.4 \text{ ha/hr}}$$

$$\text{Eff} = 86.25\%$$



Under load rotates at 1200 rpm and at no load, 1125 rpm. What is the percent slippage of tractor drive wheel?

- a. 6.80%
- b. 6.77%
- c. 6.67%
- d. 6.25%


$$\% \text{ Slippage} = \frac{N_{\text{under load}} - N_{\text{no load}}}{N_{\text{under load}}} = \frac{D_{\text{actual}} - D_{\text{theoretical}}}{D_{\text{theoretical}}}$$

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$$\% \text{ Slippage} = \frac{1200 \text{ rpm} - 1125 \text{ rpm}}{1200 \text{ rpm}}$$

$$\% \text{ Slippage} = 6.25\%$$



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- a. 6.80%
- b. 6.77%
- c. 6.67%
- d. 6.25%**

A 15-meter wheel axle hp tractor is running at a speed of 10 kph during harrowing in a deep mud soil. The tractor weighs 500 kg. The wheel diameter of the tractor is 2 m and is able to rotate at 14 revolutions in 100 m distance. The rolling resistance is 0.25 rolling coefficient.

- a. 12.56%
- b. 13.77%
- c. 13.68%
- d. 12.43%

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$$\% \text{ Slippage} = \frac{D_{actual} - D_{theoretical}}{D_{theoretical}}$$

$$\% \text{ Slippage} = \frac{100 \text{ m} - (\pi \times 2 \text{ m} \times 14 \text{ rev})}{(\pi \times 2 \text{ m} \times 14 \text{ rev})}$$

$$\% \text{ Slippage} = 0.1368 \text{ or } 13.68\%$$

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- c. 13.68%**
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It is a type of manual weeder that uses blade and tine soil-working parts with short, medium, or long handles.

- a. Push-type weeder
- b. Hand-held weeder
- c. Rotary weeder
- d. Wheel hoe



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- a. Push-type weeder
- b. Hand-held weeder**
- c. Rotary weeder
- d. Wheel hoe

Classification of Weeder

A. By design of soil working part:

1. **Blade type** – rectangular, triangular and trapezoidal shapes with cutting edges sharpened and hardened for soil cutting and weed uprooting.
2. **Tine type** – straight or curved, round or square cross-section steel rods sharply pointed and hardened at the soil working end.
3. **Rotary type** – curved spikes or paddles attached radially to a common axle which rotate when pushed forward to uproot and bury weeds.

B. By power source:

1. **Manual weeders** – hand-held or push-type weeders for upland or lowland.
 - a) **Hand-held weeders** – utilize the blade and tine type of soil working parts with short (< 0.5 m), medium (< 1 m) and long (> 1 m) handles.
 - b) **Push-type weeders** – utilize any of the three types of soil working parts for upland (wheel hoe) and lowland (rotary weeder) weeding.

B. By power source:

- 2. Animal-drawn weeders** – soil working parts are mounted on a frame or tool bar pulled by an animal for upland weeding.
 - 3. Tractor-drawn weeders** – soil working parts are mounted on a frame or tool bar pulled by a two-wheel or four-wheel tractor for upland weeding.
 - 4. Power weeders** – rotary type weeder driven by its own engine for lowland weeding.
- 

Determine the performance index of an animal drawn weeder that has a working width of 10 inches if the weeding index is 85% and field efficiency is 80%. The speed of operation is 3 kph. During the weeding operation, a 12.5% damage was observed. Assume that weeder will be powered by animal.

- a. 449.67 ha/hp-hr
- b. 451.09 ha/hp-hr
- c. 453.49 ha/hp-hr
- d. 455.12 ha/hp-hr


$$\text{Perf. Index} = \frac{\text{FC} \times (100 - \% \text{Damage}) \times \text{Eff}}{\text{Power}}$$

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$$\text{Perf. Index} = \frac{\text{FC} \times (100 - \% \text{Damage}) \times \text{Eff}}{\text{Power}}$$

$$\text{FC} = \frac{\text{SWEff}}{10} = \frac{(3 \text{ kph})(10 \text{ inch} \times \frac{0.0254 \text{ m}}{1 \text{ inch}})(0.80)}{10} = 0.06096 \text{ ha/hr}$$

$$\text{Perf. Index} = \frac{\frac{0.06096 \text{ ha}}{\text{hr}} \times (100 - 12.5) \times 85}{1 \text{ hp}}$$

$$\text{Perf. Index} = 453.39 \text{ ha/hp-hr}$$

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- a. 449.67 ha/hp-hr
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- d. 455.12 ha/hp-hr

A field was conducted to determine the efficiency of 2-wheel tractor with rototiller as implement. The following data were obtained:

Test Area = 25 m x 20 m

Rototiller width of cut = 4 ft

Total time of testing = 25 minutes

No. of turns from head land to head land = 8 turns

Speed = 40 m/min

Calculate the percent unrototilled.

a. 1.05% b. 2.44% c. 3.19% d. 4.835

Test Area = 25 m x 20 m

Rototiller width of cut = 4 ft x $\frac{1 \text{ m}}{3.28 \text{ ft}}$ = 1.2195 m

Total time of testing = 25 minutes

No. of turns from head land to head land = 8 turns

Speed = 40 m/min

Calculating rototilled area (A1):

A1 = Width of cut x No. of turns x Headland travelled x Length of plot

A1 = 1.2195 m x 8 turns x 2 headland/turn x 25 m/headland = 487.8 m²

Calculating total area (A2):

A2 = 25 m x 20 m = 500 m²

Calculating Percent Rototilled:

%Rototilled = A1/A2 = (487.8 m²/500 m²) x 100 = 97.56%

%Unrototilled = 100% - 97.56% = **2.44%**

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Process of separating the grains from the straw.

- a. Harvesting
- b. Separation
- c. Threshing
- d. Cleaning



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Harvesting – process of gathering the useful portion of the crop from the field

Threshing – process of separating the grains from the straw. Additionally, it may include cleaning the grains.



How many man-hours does it need to do manual harvesting for a hectare of field?

- a. 0–80 hrs
- b. 80–160 hrs
- c. 160–240 hrs
- d. All of the above



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In this system, harvesting is done manually while threshing is carried out mechanically.

- a. Traditional
- b. Conventional
- c. Mechanical
- d. Combined



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- a. Traditional
- b. Conventional**
- c. Mechanical
- d. Combined

SYSTEM OF HARVESTING AND THRESHING

1. Traditional system

Harvesting – done manually

- The traditional method requires the worker to hold the straw with one hand while cutting it with the other. After cutting, the harvested crop is either left in the field or stacked into “mandalas” or haystacks.
- Manual harvesting typically takes about **80 to 160 man-hours/hectare**.

Threshing – done manually

- Involves the use of non-mechanical power sources to separate the grains from the straw.
- **Hampasan** is a popular device for farmers who used it by striking the panicles to wooden slats
- Traditional threshing typically takes about **140 to 220 man-hours/hectare**

SYSTEM OF HARVESTING AND THRESHING

2. Conventional System – most commonly adopted system

- Harvesting – done manually
- Threshing – done using mechanical thresher

3. Mechanical System

- Harvesting – done mechanically
- Threshing – done mechanically

4. Combine harvesting – both operations done by a single machine – combine

5. Stripping harvesting – removal of grains from panicle without cutting the straw – stripper



The panicle is made to pass between two bars of "V" configuration to separate the grains from the straw.

- a. Shearing
- b. Stripping
- c. Separating
- d. Barring



The panicle is made to pass between two bars of "V" configuration to separate the grains from the straw.

- a. Shearing
- b. Stripping**
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- d. Barring



COMBINE STRIPPER

1. Separating the rows
2. Cutting the standing crop
3. Conveying the cut materials into the threshing section
4. Threshing
5. Shaking the straw to separate loose grains from the straw
6. Cleaning the grains of chaff
7. Conveying the grains to the tank or sack



STRIPPER

1. Separating the rows
2. Stripping
3. Cleaning the grains
4. Conveying the grains to the tank or sacks

Rice combine harvester – Mobile machine that cuts, threshes, separates, cleans, and conveys grain into a hopper or bag, while leaving residue on the ground.

PARTS OF RICE COMBINE HARVESTER

- **Oscillating screen** – Cleaning screen vibrating 300–400 times per minute.
- **Reel** – Rotating arms with battens that hold and guide the crop to the cutter and feeder.
- **Shaker shoe/shoe** – Oscillating frame supporting cleaning sieves and chaffer.
- **Straw walker** – Two or more racks that agitate straw to separate remaining grains.
- **Stripper beater / Rear beater** – Component behind the cylinder that redirects straw onto the straw walker.
- **Threshing cylinder / Drum** – Rotating assembly with rasp bars or spikes, working with a stationary element to thresh the crop.

CLASSIFICATION OF RICE COMBINE HARVESTER

Ride-on – Type of rice combine harvester where the engine is integral with the machine and provides power for all processes and movement.

Wheeled type – Rice combine harvester in which the pneumatic wheels or steel cage wheels are used

Track type (Crawler type) – Rice combine harvester fitted with full or half tracks instead of pneumatic wheels

Walk-behind – Smaller rice combine harvester wherein the engine is integral within the machine and is operated solely by one operator

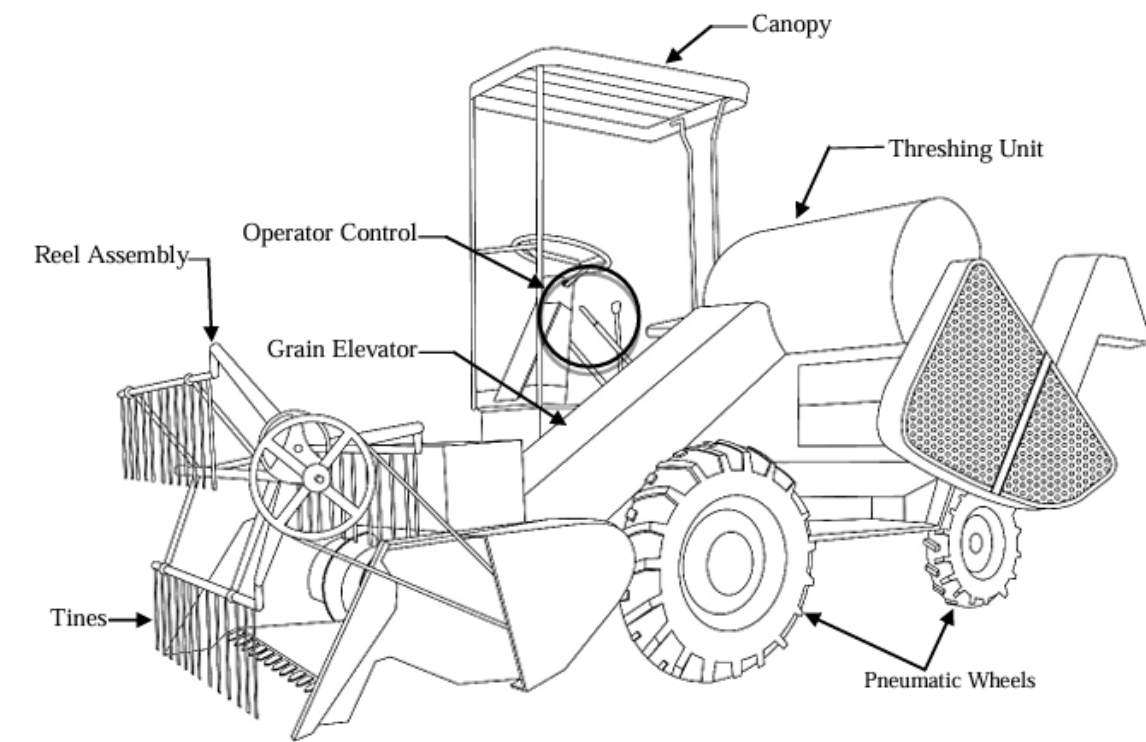


Figure 1 -Wheeled type

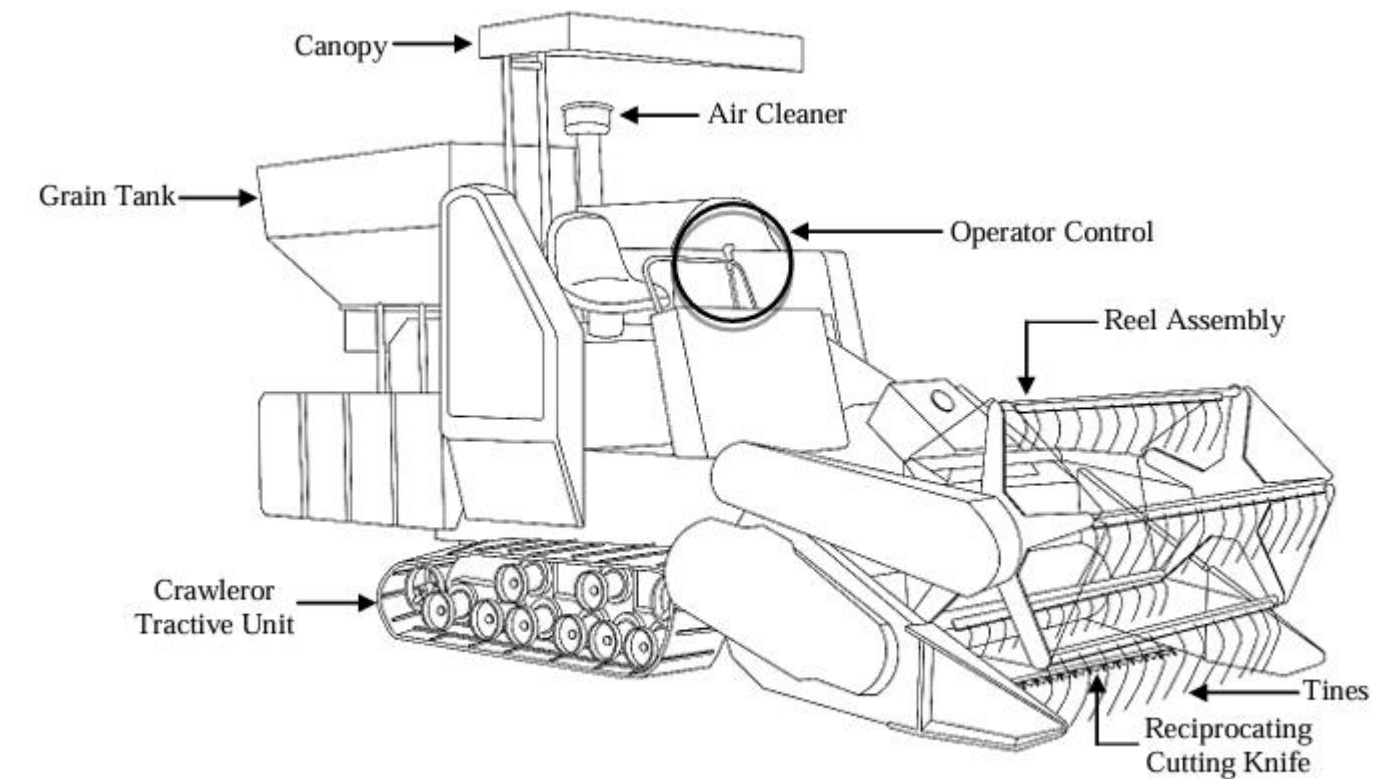


Figure 2- Track type rice combine harvester



What is the field efficiency of rice combine harvester?

- a. 75%
- b. 80%
- c. 85%
- d. 90%



What is the field efficiency of rice combine harvester?

- a. 75%
- b. 80%
- c. 85%
- d. 90%

Table 1 - Performance criteria for rice combine harvester

Criteria	Performance Data
Field efficiency (%), minimum	75.0
Header loss (%), maximum	0.5
Separation loss, (%), maximum	1.3
Blower loss, (%), maximum	1.2
Unthreshed loss, (%), maximum	0.2
Scattering loss, (%), maximum	0.3
Purity (%), minimum	
a. with fan	97
b. without fan	95
Broken grains (%), maximum	2.0
Net cracked grain (%), maximum	5.0
Noise level, db(A), maximum	95.0*

*Allowable noise level for four (4) hours of continuous exposure based on Occupational Safety and Health Standards, Department of Labor and Employment, Philippines. 2013.



What is the major factor affecting choice of machine?

- a. Farm size
- b. Kind of crop
- c. Timeliness of operation
- d. Availability of capital



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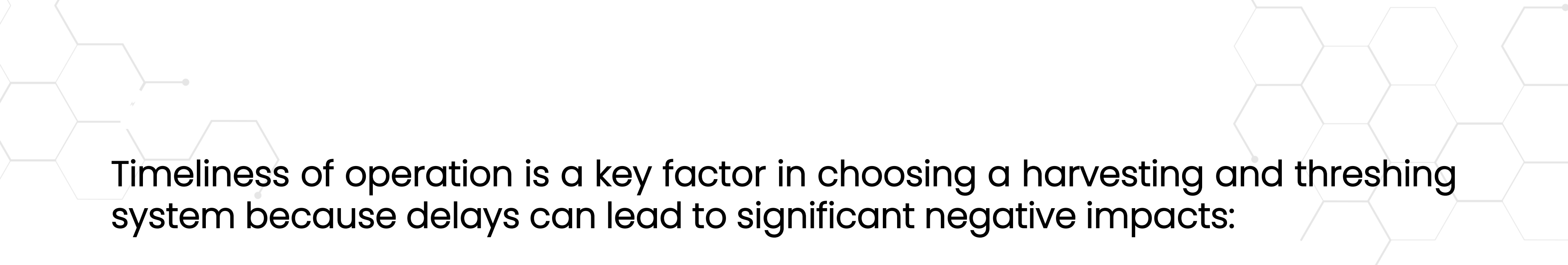


FACTORS AFFECTING CHOICE OF SYSTEM



1. Kind of crop
2. Timeliness of operation
3. Topography
4. Farm size
5. Type of culture (row or broadcasted, upland or lowland)
6. Availability and cost of labor
7. Availability of capital

A major factor affecting choice of system is **timeliness of operation** because it affects field losses, time available for the next crop and grain quality



Timeliness of operation is a key factor in choosing a harvesting and threshing system because delays can lead to significant negative impacts:

- a) **Field losses** – If harvesting is delayed, crops may shatter, lodge, or get damaged by pests and weather, leading to higher grain losses.
- b) **Time for the next crop** – In cropping systems with limited windows (e.g., rice–wheat rotations), late harvesting reduces the time for land preparation and planting the next crop, affecting overall farm productivity.
- c) **Grain quality** – Delays can cause grains to over-dry, discolor, or sprout, reducing their market value and suitability for storage.



A farmer is preparing for rice harvest. To make the soil firm enough for easier harvesting and machine movement, when should the field be drained before harvesting?

- a. 2–3 days before harvesting
- b. 1–1.5 weeks before harvesting
- c. At the time of harvesting
- d. 3–4 weeks before harvesting



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- c. At the time of harvesting
- d. 3–4 weeks before harvesting



A machine that cuts the standing crop, conveys the cut crop to one side, and lays them down in an orderly manner.

- a. Hand tools
- b. Reapers-windrower
- c. Reaper-binder
- d. None of the above



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- a. Hand tools
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- c. Reaper-binder
- d. None of the above



GROUPINGS OF HARVESTERS



1. **Hand tools** – include the sickle, scythe, yatab, lingkao and cradle
2. **Reapers-windrower** – a machine that cuts the standing crop, conveys the cut crop to one side, and lays them down in an orderly manner.
3. **Reaper-binder** – a machine that cuts the standing crop, binds the cut crop, and lays down the bound crop to one side



What mechanism is employed in sickels?

- a. Slicing/tearing action
- b. impact action
- c. scissor type action
- d. Two of the choices above



What mechanism is employed in sickels?

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What mechanism is employed in combines?

- a. Slicing/tearing action
- b. impact action
- c. scissor type action
- d. Two of the choices above



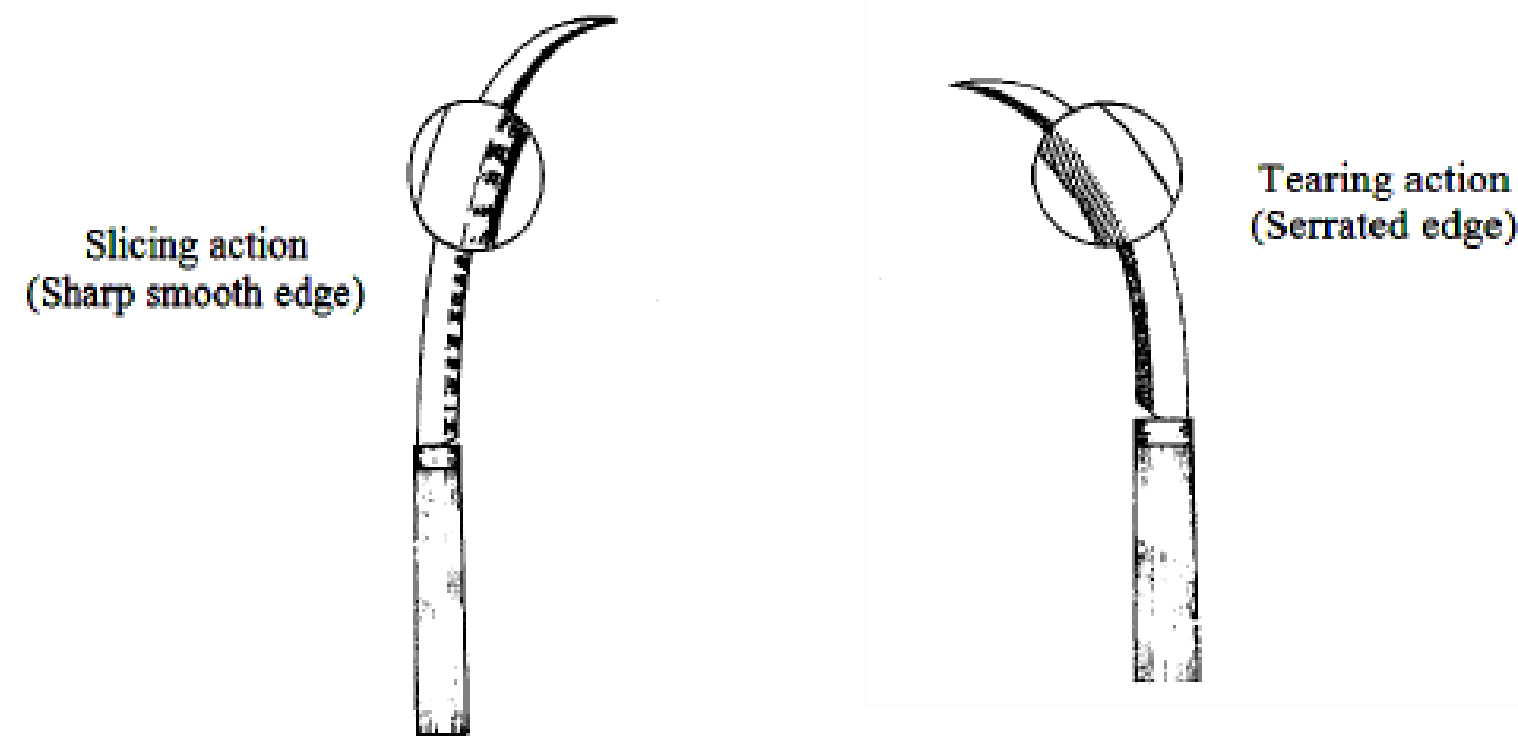
What mechanism is employed in combines?

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CUTTING MECHANISMS

Slicing/tearing action

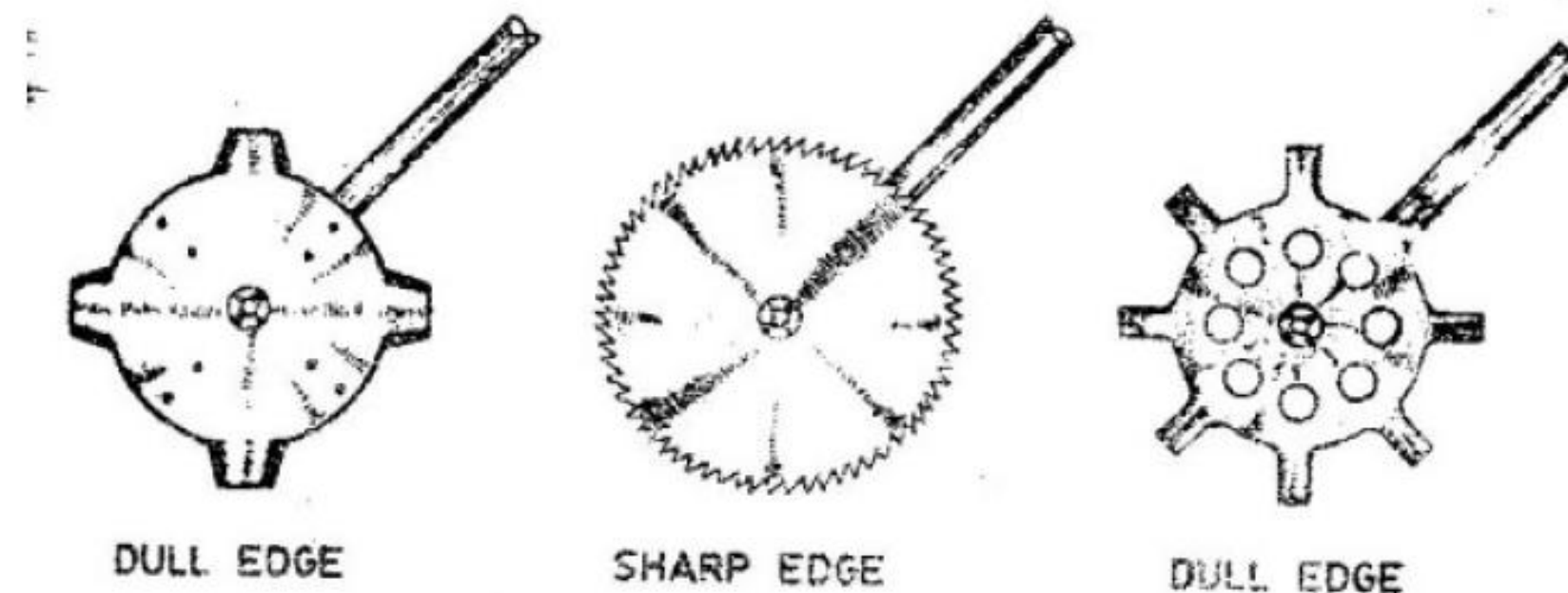
- Employed in hand tools like sickles
- Slicing action – use of sharp, smooth-edged blades
- Tearing action – use of rough serrated edged blades



CUTTING MECHANISMS

High velocity, single-element, impact action

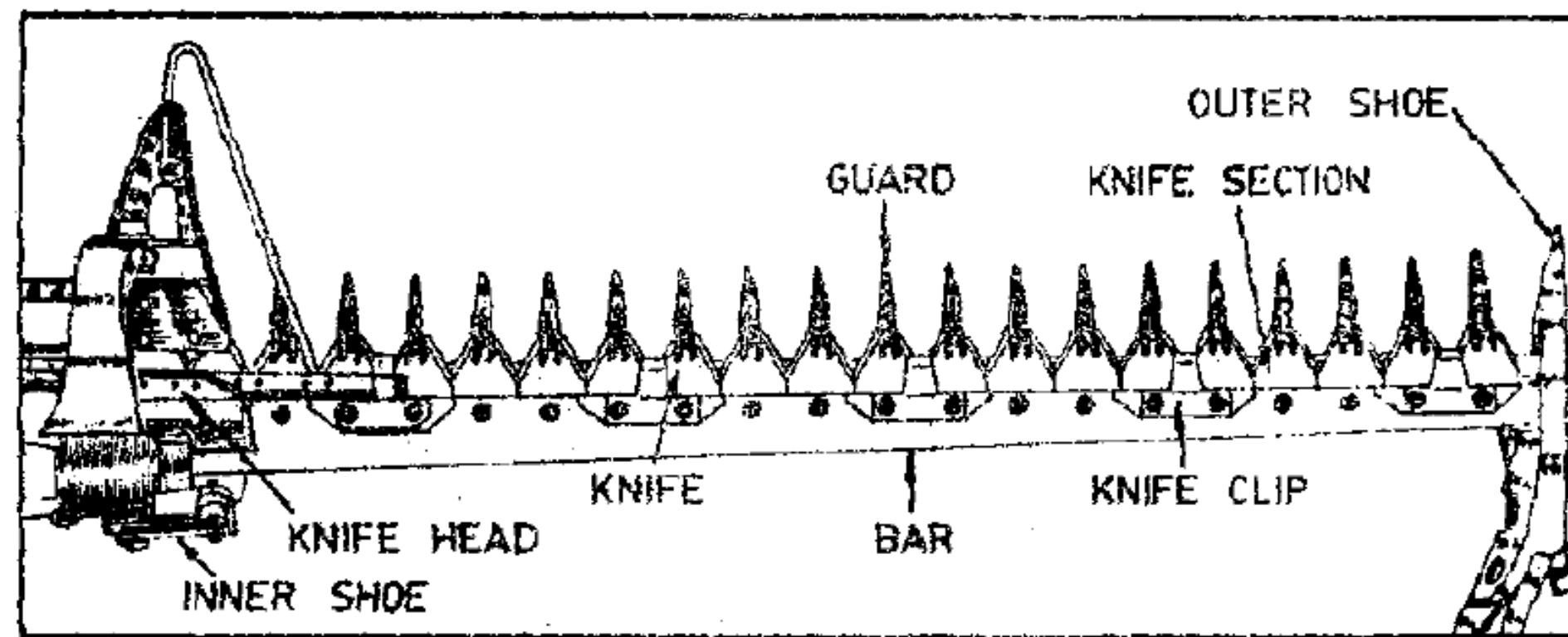
- use of sharp or dull-edged blades moving at high velocity of 2,000 fpm to 9,000 fpm
- Portable power scythes and some reapers use this method



CUTTING MECHANISMS

Two-element, scissor type action

- Shearing action between the moving and stationary blades
- Most common cutting mechanism used for reapers and combines





The output of man treading is _____.

- a. 34 kg/hr
- b. 24 kg/hr
- c. 14 kg/hr
- d. None of the above



The output of man treading is _____.

- a. 34 kg/hr
- b. 24 kg/hr
- c. 14 kg/hr
- d. None of the above



METHODS OF THRESHING



1. **Rubbing action**– grains are detached from their panicles because of a rubbing action as in treading by man, animal and vehicle.
 - Output of man treading is 14 kg/hr
2. **Impact action** – grains are accelerated faster than their panicles and are detached as in hampasan and mechanical threshers.
 - Output of hampasan is 34 kg/hr
3. **Stripping action** – grains are detached from their panicles when the straw is pulled through a “V” configuration or a comb-like device is passed through the panicles.



What mechanism is employed in mechanical thresher?

- a. Slicing/tearing action
- b. impact action
- c. scissor type action
- d. Two of the choices above



What mechanism is employed in mechanical thresher?

- a. Slicing/tearing action
- b. impact action**
- c. scissor type action
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What threshing cylinder is commonly used by axial flow threshers for rice?

- a. Wire loops
- b. Peg
- c. Spikes
- d. Rasp-bar



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- b. Peg
- c. Spikes
- d. Rasp-bar

MECHANICAL THRESHER

1. Power source – manual as in pedal thresher or power thresher as in engine-driven thresher.

2. Type of feeding

a. Hold-on feeding – Straws do not pass through the threshing section

- Examples: Pedal thresher and Japanese combine

b. Throw-in feeding – Straws pass through the threshing section

- Examples: Axial-flow thresher and US combine

3. Direction of threshing materials

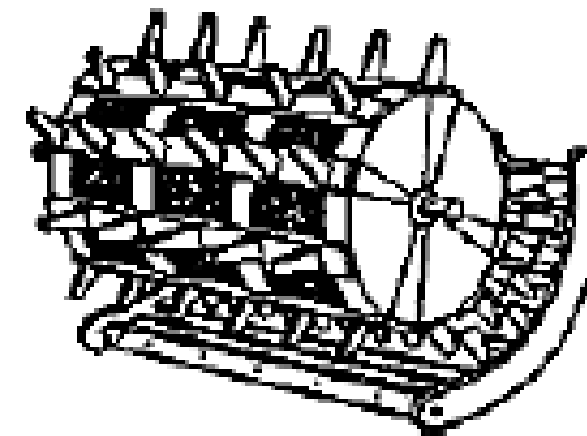
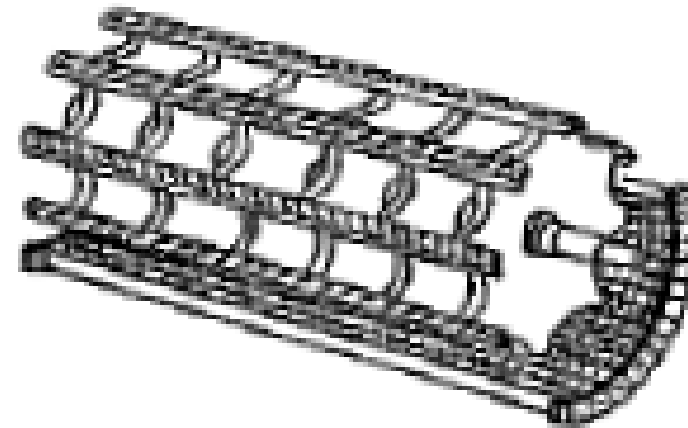
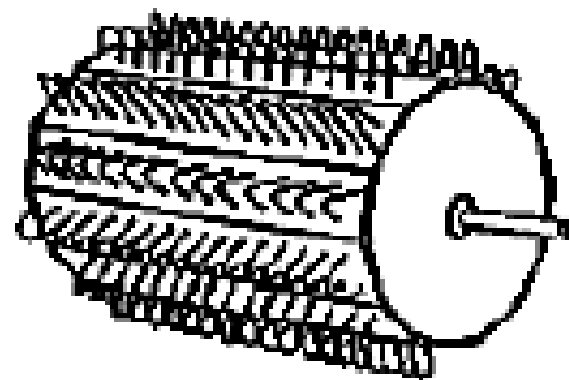
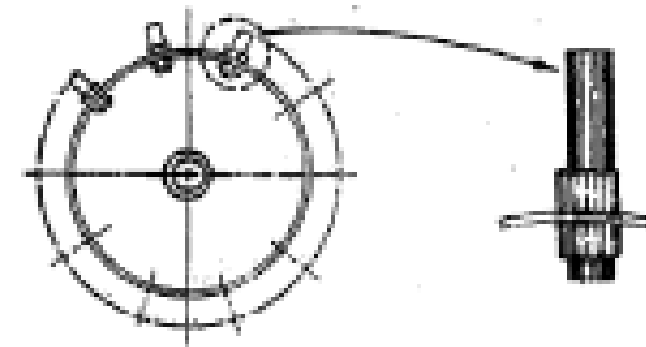
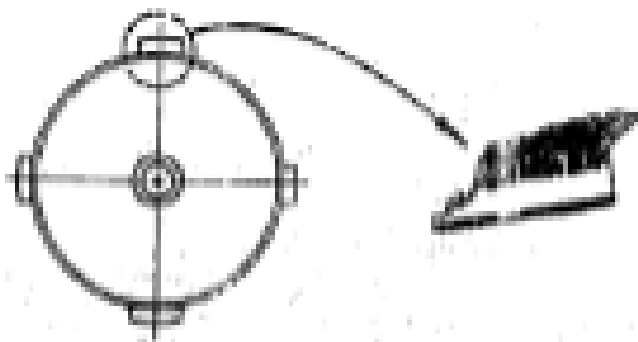
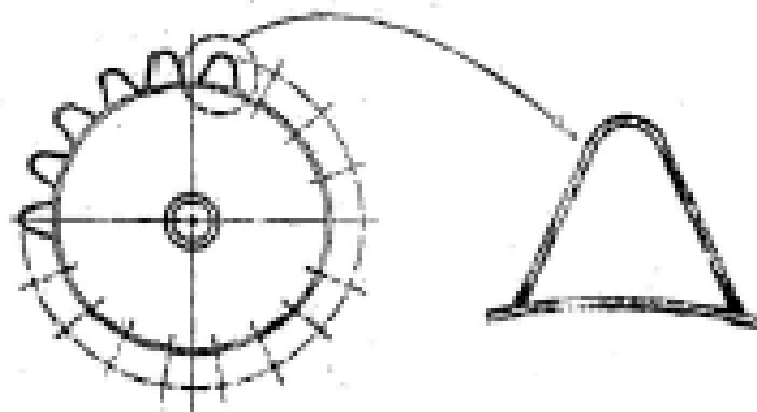
a. Tangential-flow – Crop passes between the cylinder and concave, exiting tangentially (About 60% of the grains pass through the concave)

b. Axial-flow – Crop enters one end, circles the cylinder axially several times, and exits the other end. (About 90% of the grains are separated from the straw at the cylinder.)

MECHANICAL THRESHER

4. Type of cylinder teeth

- a. **Wire loop** – equipped with wire loops
- b. **Rasp-bar cylinder** – bar-like protrusions in parallel orientation; adopted for wide variety of crops like peanut and other cereals because of its mild action
- c. **Peg-toothed cylinder** – equipped with spikes or pegs; most common used by axial flow threshers for rice





Which of the following components is correctly matched with its corresponding part in a BLOWBAGETS inspection?

- a. W – Check for tire pressure and alignment.
- b. L – Ensure headlights and warning lights are functioning.
- c. O – Inspect the fuel gauge for sufficient gas.
- d. S – Verify that all hydraulic systems are working properly.



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BLOWBAGETS



BRAKES



LIGHTS



OIL



WATER



BATTERY



AIR



GAS



ENGINE



TIRES



SELF



In a routine check before using the tractor, what should be checked first?

- A. Fuel
- B. Water
- C. Brake
- D. Oil



In a routine check before using the tractor, what should be checked first?

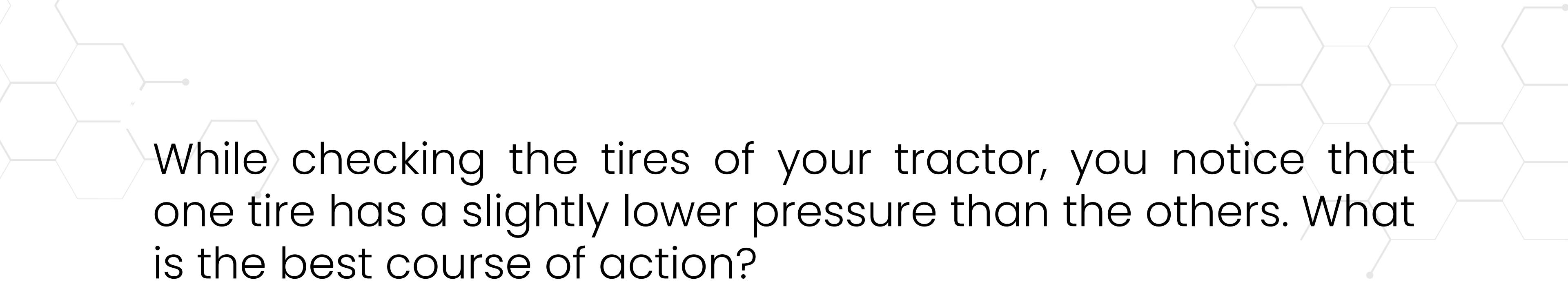
- A. Fuel
- B. Water
- C. Brake**
- D. Oil

Before using a tractor for the day, you perform a BLOWBAGETS inspection. You notice that the water level in the radiator is below the recommended level. What should you do?

- a. Continue operating the tractor since the water will circulate when the engine heats up.
- b. Add cold water immediately, regardless of engine temperature.
- c. Turn off the tractor, let the engine cool, then refill the radiator.
- d. Reduce the workload on the tractor to avoid overheating.

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- d. Reduce the workload on the tractor to avoid overheating.



While checking the tires of your tractor, you notice that one tire has a slightly lower pressure than the others. What is the best course of action?

- a. Inflate the tire until it feels firm by hand pressure.
- b. Inflate the tire to the manufacturer's recommended PSI level.
- c. Ignore the issue if the difference is minor.
- d. Only increase the pressure if you plan to drive on paved surfaces.



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During your tractor inspection, you find that the brake fluid level is lower than normal. What should you do?

- a. Top up the fluid and continue operation.
- b. Ignore the issue if the brakes still feel responsive.
- c. Inspect for potential leaks before refilling the fluid.
- d. Avoid using the brakes to conserve fluid until repairs can be made.



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- a. Top up the fluid and continue operation.
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You are about to park your tractor after completing fieldwork. Which of the following options correctly represents the proper sequence for stopping the tractor?

- I. Apply brakes to stop the tractor.
- II. Disengage the clutch by pressing the clutch pedal and shifting to neutral.
- III. Reduce engine speed to idling using the hand throttle.
- IV. Engage the parking brake and dismount safely from the left side.
- V. Turn off the engine using the fuel shut-off knob/stop switch and remove the key.

a. I, II, III, IV, V

b. III, II, I, V, IV

c. III, I, V, IV, II

d. II, I, IV, III, V

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a. I, II, III, IV, V

b. III, II, I, V, IV

c. III, I, V, IV, II

d. II, I, IV, III, V

STEPS FOR STOPPING A TRACTOR

1. Reduce the engine speed (by hand throttle lever) to idling position.
2. Press the clutch pedal to disengage the clutch and put the gear shift lever in neutral position.
3. Release the clutch.
4. Stop the tractor (by applying brakes)
5. Pull the fuel shut-off knob/ stop switch till engine stops.
6. Withdraw key by turning it anticlockwise.
7. Engage parking brake.
8. Get up from the seat
9. Get down from the tractor from left side only.



The time rate of doing work or delivering energy, expressed as the amount of work W done or energy transferred divided by the time elapsed t , computed as W/t , is called:

- A. Force
- B. Power
- C. Work
- D. Energy



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- A. Force
- B. Power**
- C. Work
- D. Energy



What is the nominal battery voltage of a 4-wheel, 90-hp tractor?

- A. 12 Vac
- B. 24 Vdc
- C. 9 Vdc
- D. 12 Vdc



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- A. 12 Vac
- B. 24 Vdc
- C. 9 Vdc
- D. 12 Vdc



A 24-volt combine harvester battery needs replacement. Two 12-volt batteries will be used. How should they be connected?

- A. Series
- B. Parallel
- C. Series–Parallel
- D. Any connection



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- A. Series
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Series vs. Parallel Battery Connection

Series

- Voltage adds, current stays the same (used for 24 V from two 12 V batteries)

Parallel

- Voltage stays the same, current capacity increases (used for jump starting same voltage systems)



The 12-volt battery of a boom sprayer is drained. Another 12-volt vehicle battery is available, with jumper wires. How should the batteries be connected for starting?

- A. Series
- B. Parallel
- C. Series–Parallel
- D. Any connection



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- A. Series
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- C. Series–Parallel
- D. Any connection



A 24-volt battery of a pineapple harvesting conveyor needs charging in the field using 12-volt solar modules. What is the minimum number of modules required?

- A. One
- B. Two
- C. Three
- D. Four



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- A. One
- B. Two**
- C. Three
- D. Four



A 12-volt boom sprayer battery needs charging in the field using 6-volt solar modules, with 2 modules in parallel to speed up charging. How many solar modules are needed?

- A. Two
- B. Four
- C. Six
- D. Eight



A 12-volt boom sprayer battery needs charging in the field using 6-volt solar modules, with 2 modules in parallel to speed up charging. How many solar modules are needed?

- A. Two
- B. Four**
- C. Six
- D. Eight



A tractor cannot start due to defective battery. It is being pushed by workers while lever is set to first gear. What should be foot-pressed and released abruptly to start engine?

- A. Brake
- B. Clutch
- C. Clutch and Brake
- D. Accelerator



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Transmission System of Four-wheel Tractor



The transmission system is consists of several speed reductions

First speed reductions – uses the pulleys and belt with a third pulley for clutching

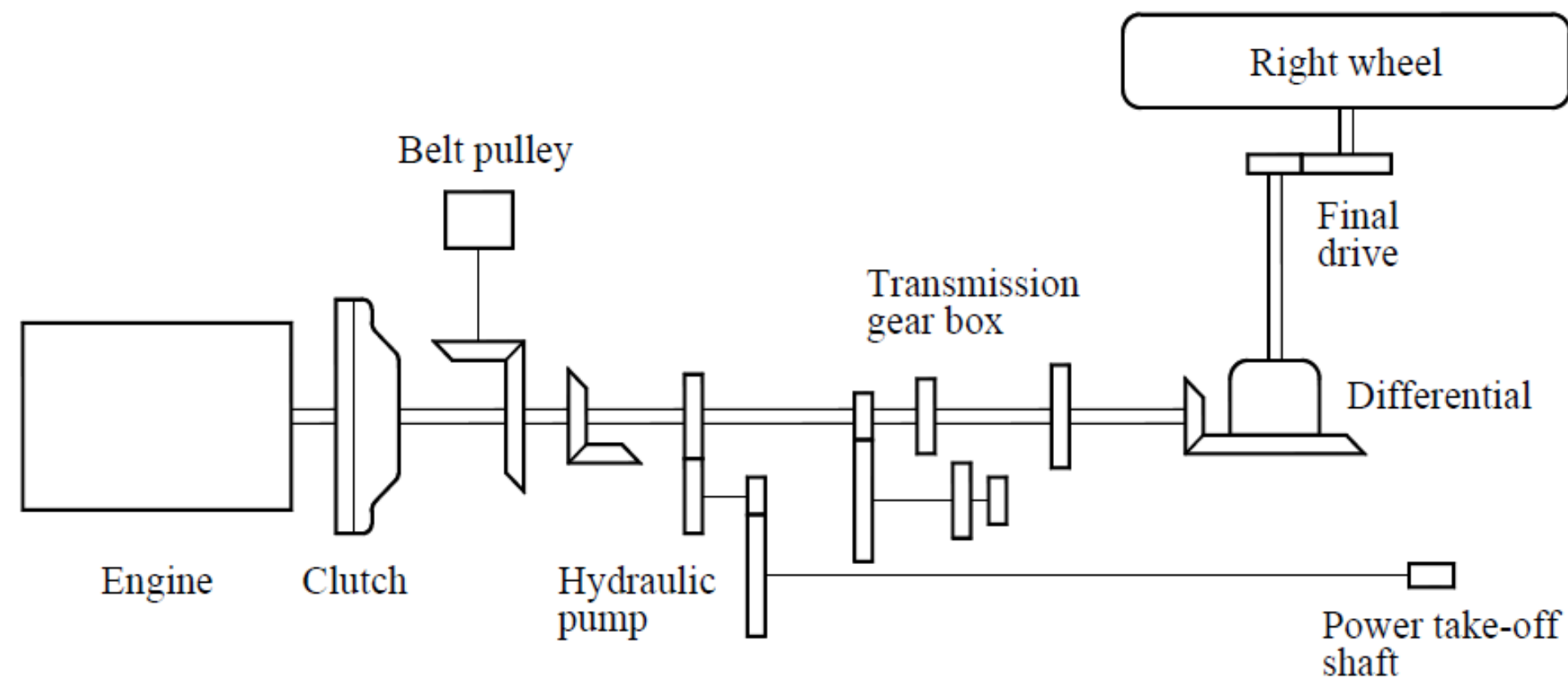
Second and third speed reductions – use the chain and sprockets enclosed in a transmission box for lubrications

RPM and diameter of the wheels – determine the linear speed of tractor

Transmission System of Four-wheel Tractor

Clutch – used to connect or disconnect the load from the engine.

Friction clutch – most common type of clutch in tractors; has a coupler that utilizes the friction between two surfaces pressed together to transmit power



Tractor Starting & Driving Basics

Push-starting

- Press clutch, get tractor moving, then release clutch abruptly to engage engine

Moving from stop

- Slowly release clutch while pressing accelerator

Slowing down

- Press brake (clutch only if stopping completely or to prevent stalling)

Stopping

- Brake first, clutch if stopping engine from stalling

Gear Selection for Safety

Descending slopes – Use low gear (first gear) for engine braking

Flat terrain starting – Lower gears prevent stalling



To gradually move a tractor from stop, what should be released while pressing the accelerator?

- A. Brake
- B. Clutch
- C. Clutch and Brake
- D. Lever



To gradually move a tractor from stop, what should be released while pressing the accelerator?

- A. Brake
- B. Clutch**
- C. Clutch and Brake
- D. Lever



To roll the tractor set at first gear in a flat land, what should be foot-pressed?

- A. Brake
- B. Clutch
- C. Clutch and Brake
- D. Accelerator



To roll the tractor set at first gear in a flat land, what should be foot-pressed?

- A. Brake
- B. Clutch
- C. Clutch and Brake
- D. Accelerator



To slow down a tractor running in first gear on flat land, what should be foot-pressed?

- A. Brake
- B. Clutch
- C. Clutch and Brake
- D. Accelerator



To slow down a tractor running in first gear on flat land, what should be foot-pressed?

- A. Brake
- B. Clutch**
- C. Clutch and Brake
- D. Accelerator



To gradually stop a fast-running tractor, what should be foot-pressed?

- A. Brake
- B. Clutch
- C. Clutch and Brake
- D. Accelerator



To gradually stop a fast-running tractor, what should be foot-pressed?

- A. Brake
- B. Clutch
- C. Clutch and Brake**
- D. Accelerator



What is the safest gear setting for a tractor descending a 10% slope?

- A. Neutral
- B. Third gear
- C. Second gear
- D. First gear



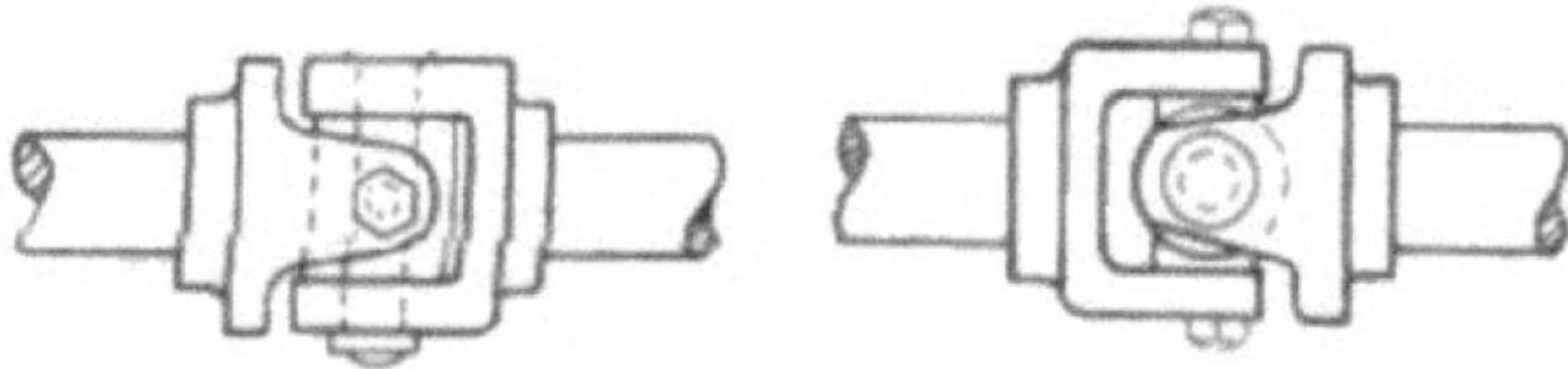
What is the safest gear setting for a tractor descending a 10% slope?

- A. Neutral
- B. Third gear
- C. Second gear
- D. First gear

Common Method of Power Transmission

Direct coupling

- Transmission efficiency = 100%
- Rotation of engine shaft = rotation of input shaft ($N_1 = N_2$)
- Direction of rotation are the same



Common Method of Power Transmission

Belt and Pulleys

- transmission efficiency = < 100% due to slippage between belt and pulley
- directions of rotation of the two pulleys are the same

Transmission efficiency is a function of belt tightness, diameter and class of pulleys, type of belt magnitude of power being transmitted

$$\frac{N_1 D_1}{(Driven)} = \frac{N_2 D_2}{(Driver)} = \frac{N_1}{N_2} = \frac{D_2}{D_1}$$

D = diameter of pulley

N = rpm of pulley



Length of Flat or V Belt

$$L = 2C + 1.57(D_1 + D_2) + \frac{(D_1 - D_2)^2}{4C}$$

Length of Cross Belt

$$L = 2C + 1.57(D_1 + D_2) + \frac{(D_1 + D_2)^2}{4C}$$

Where:

C = center-to-center distance between shaft

D1 = diameter of larger pulley

D2 = diameter of smaller pulley
(same units)

A tractor with a 9in pulley is belted to pony brake having a 24in pulley. If the tractor engine is 950rpm, the brake arm is 54in, and the net load is 60lb, what is the speed of the prony brake?

- a. 113.87 rpm
- b. 246.09 rpm
- c. 356.25 rpm
- d. 405.32 rpm

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- a. 113.87 rpm
- b. 246.09 rpm
- c. 356.25 rpm
- d. 405.32 rpm

$$N_1 D_1 (\text{driven}) = N_2 D_2 (\text{driver})$$

$$\frac{N_1}{950 \text{ rpm}} = \frac{9 \text{ in}}{24 \text{ in}}$$

$$N_1 = 356.25 \text{ rpm}$$

A tractor with a 9in pulley is belted to pony brake having a 24in pulley. If the tractor engine is 950rpm, the brake arm is 54in, and the net load is 60lb, what is the horsepower developed?

- a. 15.3 hp
- b. 16.3 hp
- c. 17.3 hp
- d. 18.3 hp

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- a. 15.3 hp
- b. 16.3 hp
- c. 17.3 hp
- d. 18.3 hp

$$N_1 D_1 (\text{driven}) = N_2 D_2 (\text{driver})$$

$$P = 2\pi F L N = \frac{2\pi(60 \text{ lbs}) \left(54 \text{ in} \times \frac{1 \text{ ft}}{12 \text{ in}} \right) (356.25 \text{ rpm})}{33,000 \text{ ft} - \text{lb}/\text{min}} = 18.3 \text{ hp}$$



An engine at 2000rpm is to drive a pump to run at 1000rpm. The engine driven pulley is 15cm in diameter and there is 10% belt slippage. What should be the diameter of the pulley on the pump?

- a. 27 cm
- b. 28 cm
- c. 29 cm
- d. 30 cm



An engine at 2000rpm is to drive a pump to run at 1000rpm. The engine driven pulley is 15cm in diameter and there is 10% belt slippage. What should be the diameter of the pulley on the pump?

- a. 27 cm
- b. 28 cm
- c. 29 cm
- d. 30 cm

$$N_1 D_1 (\text{driven}) = N_2 D_2 (\text{driver})$$

$$D_1 = \frac{N_2 D_2}{N_1} = \frac{(2000 \text{ rpm})(0.9)(15 \text{ cm})}{(1000 \text{ rpm})} = 27 \text{ cm}$$



Determine the length of flat-belt pulley drive having the following data:

Diameter of first pulley = 300 mm

Diameter of second pulley = 400 mm

Center-to-center distance of pulley and shaft = 10 m

- a. 18.1 m
- b. 19.1 m
- c. 20.1 m
- d. 21.1 m

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Diameter of second pulley = 400 mm

Center-to-center distance of pulley and shaft = 10 m

a. 18.1 m

b. 19.1 m

c. 20.1 m

d. 21.1 m

$$L = 2C + 1.57(D_1 + D_2) + \frac{(D_1 - D_2)^2}{4C}$$
$$L = 2(10) + 1.57(0.4 + 0.3) + \frac{(0.4 - 0.3)^2}{4(10)}$$
$$L = 21.1 \text{ m}$$



The cylinder of a corn sheller will be driven by a 10-hp diesel engine using cross belt drive. The engine has a speed of 1800 rpm while corn sheller requires an engine speed of 650 rpm. The effective diameter of driver pulley is 12 inches. What is the require length of cross belt if center-to-center distance between pulleys is 60 in.

- a. 198.1 in
- b. 199.1 in
- c. 200.1 in
- d. 201.1 in



The cylinder of a corn sheller will be driven by a 10-hp diesel engine using cross belt drive. The engine has a speed of 1800 rpm while corn sheller requires an engine speed of 650 rpm. The effective diameter of driver pulley is 12 inches. What is the require length of cross belt if center-to-center distance between pulleys is 60 in.

- a. 198.1 in
- b. 199.1 in**
- c. 200.1 in
- d. 201.1 in

$$N_1 D_1 = N_2 D_2$$

$$650(D_1) = 1800(12)$$

$$D_1 = 33 \text{ inches}$$

$$L = 2C + 1.57(D_1 + D_2) + \frac{(D_1 + D_2)^2}{4C}$$

$$L = 2(60) + 1.57(33 + 12) + \frac{(33 + 12)^2}{4(60)}$$

$$L = 199.0875 \text{ m}$$

Common Method of Power Transmission

Chain and Sprockets

- Two sprockets are used and connected by a roller chain
- transmission efficiency = < 100% due to chain slack
- Direction of rotation of two sprockets is the same

$$N_1 T_1 = N_2 T_2 = \frac{N_1}{N_2} = \frac{T_2}{T_1}$$

T = number of teeth
N = rpm

Common Method of Power Transmission

Gear system

- Two gears are used without the need of a third element by placing them side-by-side and transmitting power through their teeth
- Transmission efficiency = $\leq 100\%$ due to gear slack
- Direction of rotation of the two gears is opposite each other

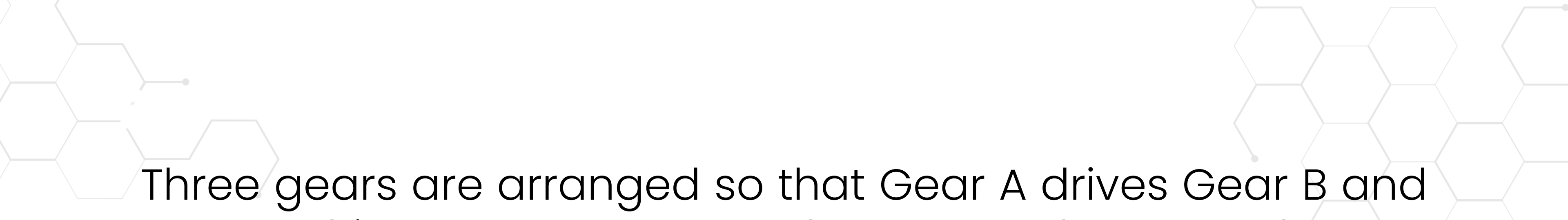
$$\text{Speed Ratio} = \frac{N_{driven}}{N_{driver}}$$



Spur gears – transmit rotatory motion between parallel shafts



Bevel gears – used to transmit motion between intersecting shafts

A decorative background element consisting of a light gray hexagonal grid pattern, partially visible at the top of the slide.

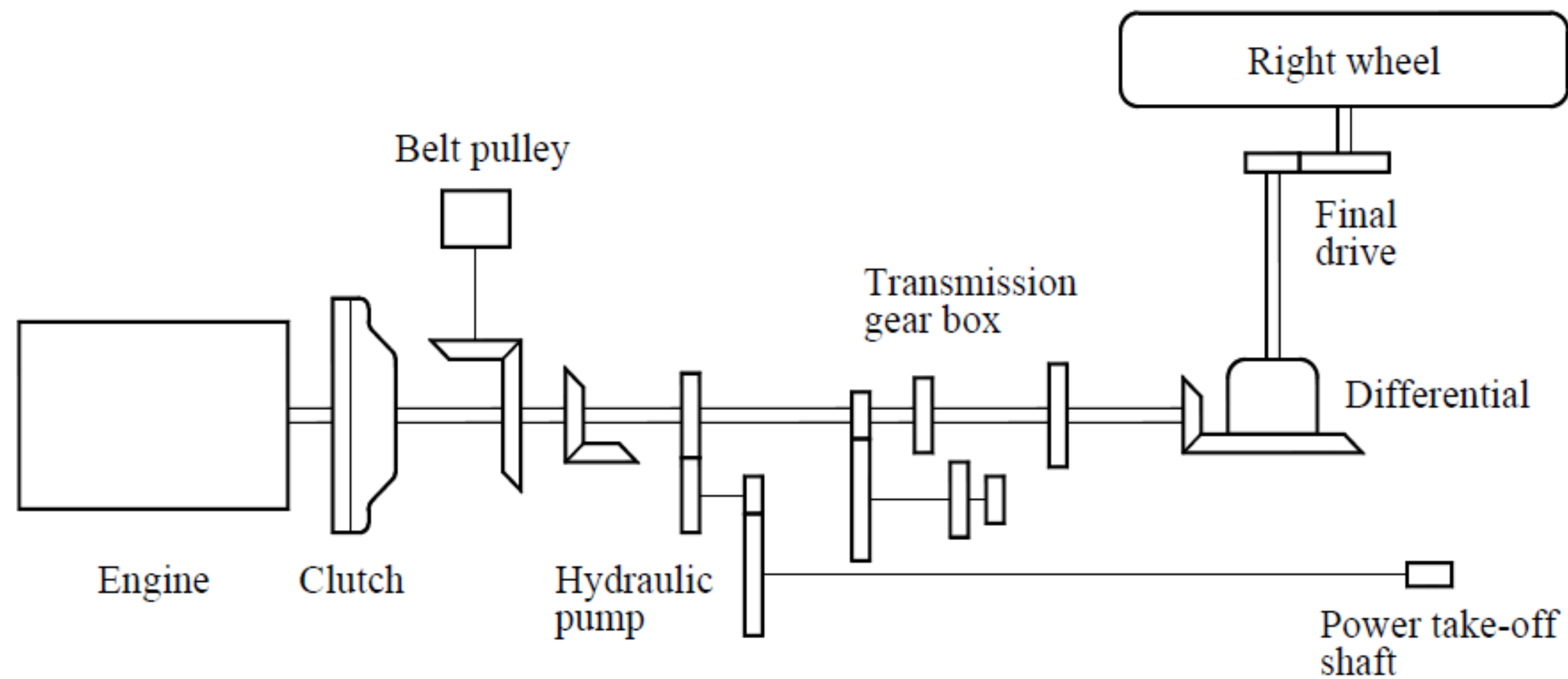
Three gears are arranged so that Gear A drives Gear B and Gear B drives Gear C. Gear A has 40 teeth, Gear B has 30 teeth and Gear C has 20 teeth. What is the overall gear ratio?

- a. 0.25
- b. 0.50
- c. 0.75
- d. 1.00

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- a. 0.25
- b. 0.50**
- c. 0.75
- d. 1.00

$$SR = \frac{N_{driven}}{N_{driver}}$$
$$SR = \frac{B}{A} \times \frac{C}{B} = \frac{30}{40} \times \frac{20}{30} = 0.5$$



$$N_1 D_1 = N_2 D_2$$

$$N_2 = N_3$$

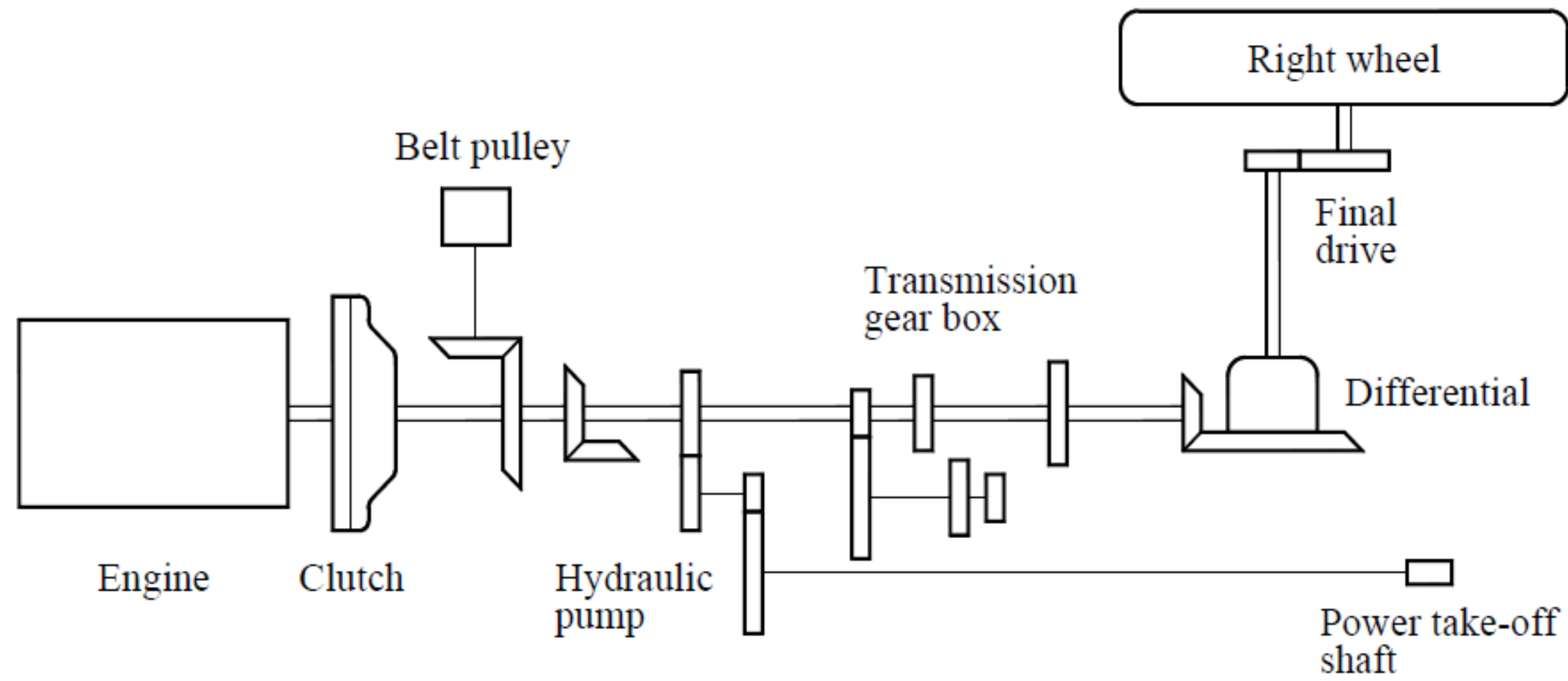
$$N_3 T_3 = N_4 T_4$$

$$N_4 = N_5$$

$$N_5 T_5 = N_6 T_6$$

$$N_6 = N_7 \text{ (Rev. of ground wheel)}$$

$$S = 2\pi N_7 = \pi D N_7$$



$$\text{Speed Ratios} = \frac{N_{engine}}{N_{groundwheel}}$$



Compute the crankshaft rear axle speed ration required to give travel speed of 4.5mph for tractor having an engine speed of 2,000rpm and rear wheels are 60in diameter.

- a. 56.14:1
- b. 60.89:1
- c. 79.33:1
- d. 82.06:1

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- b. 60.89:1
- c. 79.33:1
- d. 82.06:1

$$V (\text{tractor}) = 4.5 \text{ mph}$$

$$V (\text{engine}) = \pi DN = \pi(60 \text{ in}) \left(\frac{0.0254m}{1in} \times \frac{1km}{1000 m} \times \frac{1mile}{1.609km} \right) \left(2000 \frac{rev}{min} \right) \left(\frac{60min}{1hour} \right)$$

$$V (\text{engine}) = 357 \text{ mph}$$

$$SR = \frac{V (\text{engine})}{V (\text{tractor})} = \frac{357}{4.5} = 79.33:1$$

Determine the forward speed of the two-wheel tractor (kph) if the specifications of the transmission devices are as given below:

- a. 6 kph
- b. 7 kph
- c. 8 kph
- d. 9 kph

1 ST REDUCTION	2 ND REDUCTION	3 RD REDUCTION	TRACTION WHEELS
N1 = 3,000 rpm	N3 =	N5 =	N7 =
N2 =	N4 = 250 rpm	N6 = 75 rpm	D7 = 50 cm
D1 = 10 cm	T3 =	T5 = 33 teeth	Speed =
D2 = 30 cm	T4 = 100 teeth	T6 =	

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D2 = 30 cm	T4 = 100 teeth	T6 =	

$$N_1 D_1 = N_2 D_2 \Rightarrow N_2 = \frac{N_1 D_1}{D_2} = \frac{(3000 \text{ rpm})(10 \text{ cm})}{30 \text{ cm}} = 1000 \text{ rpm}$$

$$N_3 = N_2 = 1000 \text{ rpm}$$

$$N_3 T_3 = N_4 T_4 \Rightarrow T_3 = \frac{N_4 T_4}{N_3} = \frac{(250 \text{ rpm})(100 \text{ teeth})}{1000 \text{ rpm}} = 25 \text{ teeth}$$

$$N_5 = N_4 = 250 \text{ rpm}$$

$$N_5 T_5 = N_6 T_6 \Rightarrow T_6 = \frac{N_5 T_5}{N_6} = \frac{(250 \text{ rpm})(33 \text{ teeth})}{75 \text{ rpm}} = 110 \text{ teeth}$$

$$N_7 = N_6 = 75 \text{ rpm}$$

$$V = \pi D N_7 = \pi(0.5 \text{ M})(75 \text{ rpm}) \left(\frac{1 \text{ km}}{1000 \text{ m}} \right) \left(\frac{60 \text{ min}}{1 \text{ hour}} \right) = 7.07 \text{ kph}$$

Determine the forward speed of the two-wheel tractor (kph) if the specifications of the transmission devices are as given below:

- a. 6 kph
- b. 7 kph**
- c. 8 kph
- d. 9 kph

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What should be the number of teeth to be meshed with T3 to combine the linear speed of a 2 wheel tractor running at 2.175mph given the following data?

- a. 115
- b. 116
- c. 117
- d. 118

1 st reduction	2 nd reduction	3 rd reduction	Ground wheel diameter
N1 = 2000rpm D1 = 300mm D2 = 600mm	T3 = 13t T4 = ?	T5 = 30t T6 = 90t	D = 500mm

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$$V = \pi D N_7$$

$$\frac{2.175 \text{ miles}}{\text{hr}} \times \frac{1 \text{ hr}}{60 \text{ min}} \times \frac{1.609 \text{ km}}{1 \text{ mile}} \times \frac{1000 \text{ m}}{\text{km}} = \pi (0.5 \text{ m}) (N_6)$$

$$N_7 = 37.13 \text{ rpm}$$

$$N_6 = N_7$$

$$N_5 T_5 = N_6 T_6$$

$$N_5 T_5 = N_6 T_6 \Rightarrow N_5 = \frac{N_6 T_6}{T_5} = \frac{(37.13 \text{ rpm})(90t)}{30t} = 111.39 \text{ rpm}$$

$$N_5 = N_4 = 111.39 \text{ rpm}$$

$$N_1 D_1 = N_2 D_2 \Rightarrow N_2 = \frac{N_1 D_1}{D_2} = \frac{(2000 \text{ rpm})(300 \text{ mm})}{600 \text{ mm}} = 1000 \text{ rpm}$$

$$N_2 = N_3 = 1000 \text{ rpm}$$

$$N_3 T_3 = N_4 T_4 \Rightarrow T_4 = \frac{N_3 T_3}{N_4} = \frac{(1000 \text{ rpm})(13 \text{ teeth})}{111.39} = 116.71 \text{ or } 117 \text{ teeth}$$



What should be the number of teeth to be meshed with T3 to combine the linear speed of a 2 wheel tractor running at 2.175mph given the following data?

- a. 115
- b. 116
- c. 117
- d. 118

1 st reduction	2 nd reduction	3 rd reduction	Ground wheel diameter
N1 = 2000rpm D1 = 300mm D2 = 600mm	T3 = 13t T4 = ?	T5 = 30t T6 = 90t	D = 500mm



The connection to the tractor of an erratic battery is to be stopped quickly. What battery terminal should be disconnected first?

- A. Positive
- B. Negative
- C. Both terminals
- D. Any terminal



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Battery Safety



- Disconnect positive (+) terminal first when removing battery (prevents short circuits)
- Connect negative (–) terminal first when installing battery
- To prevent battery drain during storage, disconnect the positive terminal



A new battery is to be installed to a tractor. What battery terminal should be connected first?

- A. Positive
- B. Negative
- C. Both terminals
- D. Any terminal



A new battery is to be installed to a tractor. What battery terminal should be connected first?

- A. Positive
- B. Negative**
- C. Both terminals
- D. Any terminal



To prevent battery drain while the tractor is not in use, which terminal should be disconnected?

- A. Positive
- B. Negative
- C. Both terminals
- D. Any terminal



To prevent battery drain while the tractor is not in use, which terminal should be disconnected?

- A. Positive
- B. Negative
- C. Both terminals
- D. Any terminal



What kind of hitch is best for a non-hydraulic disk plow?

- A. 1-point hitch
- B. 2-point hitch
- C. 3-point hitch
- D. 4-point hitch



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What kind of hitch is best for a tractor trailer?

- A. 1-point hitch
- B. 2-point hitch
- C. 3-point hitch
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- A. 1-point hitch
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You encounter an issue with a tractor that won't start. Following the basic steps of troubleshooting, what is the correct order of actions to resolve the problem?

- a. Troubleshooting → Check → Analysis
- b. Check → Troubleshooting → Analysis
- c. Analysis → Check → Troubleshooting
- d. Check → Analysis → Troubleshooting



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- c. Analysis → Check → Troubleshooting
- d. Check → Analysis → Troubleshooting



PROCESS OF TROUBLESHOOTING



1. **Analysis** – Start by understanding potential causes based on symptoms or prior knowledge.
2. **Check** – Inspect the tractor's components systematically (e.g., battery, fuel, air filters) to confirm or rule out suspected issues.
3. **Troubleshooting** – After confirming the cause, take corrective action to resolve the problem.
4. This sequence ensures a structured approach by using analysis to narrow down possibilities, checks to verify assumptions, and troubleshooting to implement solutions effectively.

5S OF HOUSEKEEPING TRACTORS

HOUSEKEEPING	EXAMPLES	PURPOSE
1. Sort (Seiri) - Remove unnecessary items from the tractor and its surroundings	Get rid of expired lubricants, broken tools, or unused attachments.	Reduces clutter, making maintenance more efficient and the workspace safer.
2. Set in Order/Systemized (Seiton) - Organize tools, parts, and equipment needed for maintenance in designated areas.	Store wrenches, oil cans, and replacement parts near the tractor in a systematic way.	Ensures quick access to needed tools, reducing downtime during repairs.
3. Shine (Seiso) - Clean the tractor regularly to prevent dirt buildup	Wipe down the engine, clean air filters, and wash the exterior after fieldwork.	Keeps the tractor in good condition and helps identify issues like leaks early.
4. Standardize (Seiketsu) - Create a maintenance schedule and standard procedures for inspections and servicing.	Regular oil changes, tire pressure checks, and fuel filter replacements at specified intervals.	Ensures that maintenance tasks are consistently performed to avoid breakdowns.
5. Sustain (Shitsuke) - Develop habits and discipline to maintain the 5S system consistently.	TrACTOR operators to follow cleaning routines and maintenance schedules after each use.	Ensures long-term care and reliability of the tractor, preventing neglect.



What is the main purpose of the "Sort" phase in the 5S of housekeeping when applied to tractor maintenance?

- 
- a. Organize tools and parts for quick access
 - b. Eliminate unnecessary items to reduce clutter
 - c. Ensure regular cleaning to maintain tractor health
 - d. Develop habits to sustain the system



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What is another
term for "Sustain"
phase of the 5S
method?

- 
- a. Seiketsu
 - b. Seiso
 - c. Seiri
 - d. Shitsuke



What is another
term for "Sustain"
phase of the 5S
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- 
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 - b. Seiso
 - c. Seiri
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Which of the following is an example of "Shine" in tractor maintenance?

- a. Storing wrenches and oil cans systematically
 - b. Cleaning air filters and wiping down the engine
 - c. Removing expired lubricants and broken tools
 - d. Creating a schedule for oil changes
- 



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 - c. Removing expired lubricants and broken tools
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- 



A tractor operator creates a maintenance checklist to ensure that oil changes, tire pressure checks, and air filter replacements are performed at regular intervals by all staff. Which 5S step does this represent?

- 
- a. Set in Order (Seiton)
 - b. Standardize (Seiketsu)
 - c. Shine (Seiso)
 - d. Sustain (Shitsuke)



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It is an indicative measure of the level of agricultural mechanization.

- a. Agricultural Mechanization Index
- b. Mechanization Performance Ratio
- c. Farm Power Utilization Rate
- d. Machinery Efficiency Coefficient



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Agricultural Mechanization Index

- Basis of agricultural mechanization intervention for the development of the agricultural sector
- Indicative measure of the level of agricultural mechanization



Which qualitative level of mechanization uses a combination of non-mechanical power sources and mechanical power sources operated by humans?

- a. Low Mechanization
- b. Intermediate Mechanization
- c. High Mechanization
- d. Full Mechanization



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Qualitative Levels of Mechanization



- a. **Low Mechanization** – done with use of non-mechanical power such as man and animal
- b. **Intermediate Mechanization** – done with the use of non-mechanical power source in combination with the use of a mechanical power source operated by man.
- c. **High Mechanization** – done solely with the use of mechanical power source operated by man
- d. **Full Mechanization** – done solely with the use of mechanical power source with limited intervention by man



Based on PHilMech's assessment, the highest mechanization level was logged in what region?

- 
- a. Region 2
 - b. Region 4A
 - c. BARRM
 - d. Region 13



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 - b. Region 4A
 - c. BARRM
 - d. Region 13



Based on PHilMech's assessment, the lowest mechanization level was logged in what region?

- 
- a. Calabarzon
 - b. BARRM
 - c. Mimaropa
 - d. Cordillera Administrative Region



Based on PHilMech's assessment, the lowest mechanization level was logged in what region?

- 
- a. Calabarzon
 - b. BARRM**
 - c. Mimaropa
 - d. Cordillera Administrative Region

MECHANIZATION LEVEL IN THE PHILIPPINES FROM HIGHEST TO LOWEST

Region 2: 3.5079 hp/ha
Region 4A (Calabarzon): 3.3569 hp/ha
Region 4B (Mimaropa): 2.2233 hp/ha
Region 1: 3.0936 hp/ha
Region 7: 3.019 hp/ha
Region 5: 2.9844 hp/ha
Region 6: 2.7616 hp/ha
Region 3: 2.6177 hp/ha
Region 11: 2.5912 hp/ha
Region 10: 2.2967 hp/ha
CAR: 2.3523 hp/ha
Region 8: 2.4131 hp/ha

Region 9: 1.5908 hp/ha
Region 12: 1.7217 hp/ha
Region 13: 1.6532 hp/ha
BARRM: 0.9031 hp/ha



Under the Rice Competitiveness Enhancement Fund (RCEF) Mechanization Program, how many units of various agricultural equipment has been delivered from mid-2020 to March 3, 2024 in the Philippines?

- a. 22, 844
 - b. 26,412
 - c. 30,012
 - d. 34,387
- 



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Based on the P24.9 billion funds obligated by PhilMech for the procurement of agricultural equipment, which type of agricultural equipment represents the highest proportion of the total units procured?

- 
- a. Drying technologies
 - b. Milling equipment
 - c. Harvesting and threshing equipment
 - d. Land preparation machines



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SUMMARY OF AGRICULTURAL EQUIPMENT PROCUREMENT BY PHILMECH

Total funds allocated: P24.9 billion

Total agricultural equipment: 28,817 units

- Land preparation machines: 14,713 units
- Crop establishment technologies: 4,284 units
- Harvesting and threshing equipment: 8,210 units
- Drying technologies: 768 units
- Milling equipment: 842 units

Source: PhilMech Website



What is the labor requirement
needed to operate a combine
harvester?

- 
- a. 1 person only
 - b. 1-2 persons
 - c. 2-3 persons
 - d. 2-4 persons



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SPECIFICATIONS OF
AVAILABLE TECHNOLOGIES
IN THE PHILIPPINES
FOR COMBINE HARVESTER

Source: PhilMech Website

Specifications:

Engine	65-85 hp diesel, water cooled
Drive system	Rubber crawler track
-Width	Minimum of 500 mm
Cutting system	
-Height adjustment	Hydraulic
-Width	Minimum of 1,600 mm
Output system options	Sack or bag type
Attachment	Trailer
Labor requirement	2-3 persons



What is the labor requirement
needed to operate a
thresher?

- 
- a. 1 person only
 - b. 1-2 persons
 - c. 2-3 persons
 - d. 2-4 persons



What is the labor requirement
needed to operate a
thresher?

- 
- a. 1 person only
 - b. 1-2 persons
 - c. 2-3 persons
 - d. 2-4 persons

SPECIFICATIONS OF
AVAILABLE TECHNOLOGIES
IN THE PHILIPPINES
FOR THRESHER

Source: PhilMech Website

Specifications:

Small capacity/mini rice thresher	
Engine	7-10 hp
Capacity	Minimum of 1 ton/h, 1.2 tons/h, 1.5 tons/h
Type of cylinder	Open-type cylinder, peg tooth
Threshing efficiency	Minimum of 99.8%
Component	Lifting bars
Labor requirement	2-4 persons
Large capacity	
Engine	9-15 hp
Capacity	2 tons/h



What is the labor requirement
needed to operate a reaper?

- 
- a. 1 person only
 - b. 1-2 persons
 - c. 2-3 persons
 - d. 2-4 persons



What is the labor requirement
needed to operate a reaper?

- 
- a. 1 person only
 - b. 1-2 persons
 - c. 2-3 persons
 - d. 2-4 persons

SPECIFICATIONS OF AVAILABLE TECHNOLOGIES IN THE PHILIPPINES

FOR REAPER

Source: PhilMech Website

Specifications:

Engine	3.5 hp, minimum of 4 stroke gasoline
Cutting height	Adjustable
Cutting width	1.2 m
Field efficiency	Minimum of 65 %
Components	Cage wheel, pneumatic tires
Labor requirement	1 person